

NATIONAL CHEMISTRY CONTEST CUBA 2024

ANSWERS TO THE PROBLEMS

“It should not be learned by heart only; It is necessary to learn to understand, to acquire knowledge, not to retain words; You have to learn by thinking.”

José de la Luz y Caballero



Ministry of Education
Republic of Cuba

National Chemistry Contest Cuba 2024

Answers to the problems

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“Dear contestants, we work for you because you are a source of pride for family, friends, teachers, directors and Cuban education.”

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Free of charge

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Distribution of questions by exam day and grade

	First Day	Second Day
10mo	Questions 1, 2, 3 and 4	Questions 6, 7, 8, 9 and 10
11no	Questions 1, 2, 3 and 4	Questions 11, 12, 13 and 14
12mo	Questions 3, 4, 4A and 5	Questions 13, 14, 15 and 16

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamic	$Q = \Delta U + W$
Zeroth in Celsius's scale	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs's energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Vapor pressure depression	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of the sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of the sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of the circle	$A_{\text{sp}} = \pi \cdot r^2$	Light speed	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Length of the circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity acceleration on the Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$			
$1 \text{ bar} = 100 \text{ kPa}$			
$1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$			
$1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$			
$1 \text{ dm}^3 = 1 \text{ L}$			
$1 \text{ mL} = 1 \text{ cm}^3$			
$1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3		4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																		2 He 4.003
2	3 Li 6.941	4 Be 9.012												5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305												13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956		22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906		40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	*	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207,2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	**	104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

Specific instructions for the Chemistry Contest

Dear student:

- This exam was prepared by a Scientific Committee of former Olympians who work at the University of Havana.
- **Fill in your personal details.** Make verification easy with your ID card.
- The provincial education directorate is responsible for guaranteeing a printout of the exam of sufficient quality that allows students to read the instructions, questions, figures, data, graphs, and tables. **No question will be canceled due to a wrong impression.**
- Write all your information clearly in the indicated place (names, surnames, school, municipality, province). **Please do not write anything in the CODE field;** it is for the exclusive use of the person responsible for coding.
- **Verify that your exam is complete.** This is the exam for the first day.
- 10th and 11th grades must answer 4 questions (question 1, 10 items; question 2, 10 items; question 3, 9 items; and question 4, 10 items) and the complete exam has a total of 6 pages.
- 12th grade must answer 4 questions (question 3, 9 items; question 4, 10 items; question 4A, 1 item; and question 5, 16 items) and the complete exam has a total of 7 pages.
- **Verify that your syllabus is complete.** This is the agenda for the second day of the exam.
- 10th grade must answer 5 questions (question 6, 2 paragraphs; question 7, 3 paragraphs; question 8, 3 paragraphs; question 9, 10 paragraphs; and question 10, 5 paragraphs) and the complete exam has a total of 6 pages.
- 11th grade must answer 4 questions (question 11, 5 items; question 12, 12 items; question 13, 10 items; and question 14, 12 items) and the complete exam has a total of 8 pages.
- 12th grade must answer 4 questions (question 13, 10 items; question 14, 12 items; question 15, 10 items; and question 16, 14 items) and the complete exam has a total of 9 pages.
- All necessary data, constants, conversions, equations, and periodic tables are included on the last page.
- At the beginning of each question, it indicates what grade it is directed towards, its value (out of 100 points), and the value of each section (in marks).
- You need to use a scientific calculator and a ruler.
- Follow the rules for approximate calculations.
- Attention! You have a limited time, but the exam has many sections.
- You do not have a space limit to respond. Respond in small but legible handwriting and an orderly manner.
- You must present all auxiliary calculations and **always** justify your reasoning and answers appropriately. **Answers that are not understood will not be graded.**
- The additional answer sheets at the bottom of each page indicate the total number of pages, for example, “1 of 7,” “2 of 7,” and so on.

You are a source of pride for Cuban education: Successes!

Question 1. Chemical periodicity (5% of the total; only 10th and 11th grades)

1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10	Total	Total
2	2	2	2	2	2	2	2	2	2	20	5%
2	2	2	2	2	2	2	2	2	2	20	5%

Complete the following statements.

Please do not consider the chemical elements of period 7 in your analysis since these are radioactive or artificial and, due to their high instability, their chemistry has been little studied.

1.1. The chemistry of the element Si has certain analogies with the chemistry of the element C. However, Si compounds have larger bond distances because the atomic radius of Si is larger. Furthermore, the chemical bonds of Si with F, O and N are more polar because the electronegativity of Si is lower. **2 mark**

1.2. Period 2 elements have unusually low electron affinities due to their small atomic radius and high electron density. **2 mark**

1.3. The most basic hydride of the stable alkali metals is CsH. **2 mark**

1.4. The superior oxide of the elements of group 14 with the greatest oxidizing character is PbO₂. **2 mark**

1.5. The highest secondary ionization energy in the first transition series is possessed by the element copper (Cu). **2 mark**

Chromium (Cr) is also accepted as an answer. A -1 mark is penalized if the answer is potassium.

1.6. The maximum possible oxidation state in the first transition series is not reached beyond the group VIIB/VIIIB (7/8). **2 mark**

1.7. The volatile hydrogenated compound with the highest boiling temperature among those formed by the elements of group 16 is water (H₂O). **2 mark**

1.8. The cations Cu⁺, K⁺ and Cu²⁺ are arranged in increasing order of ionic radius according to Cu²⁺ < Cu⁺ < K⁺. **2 mark**

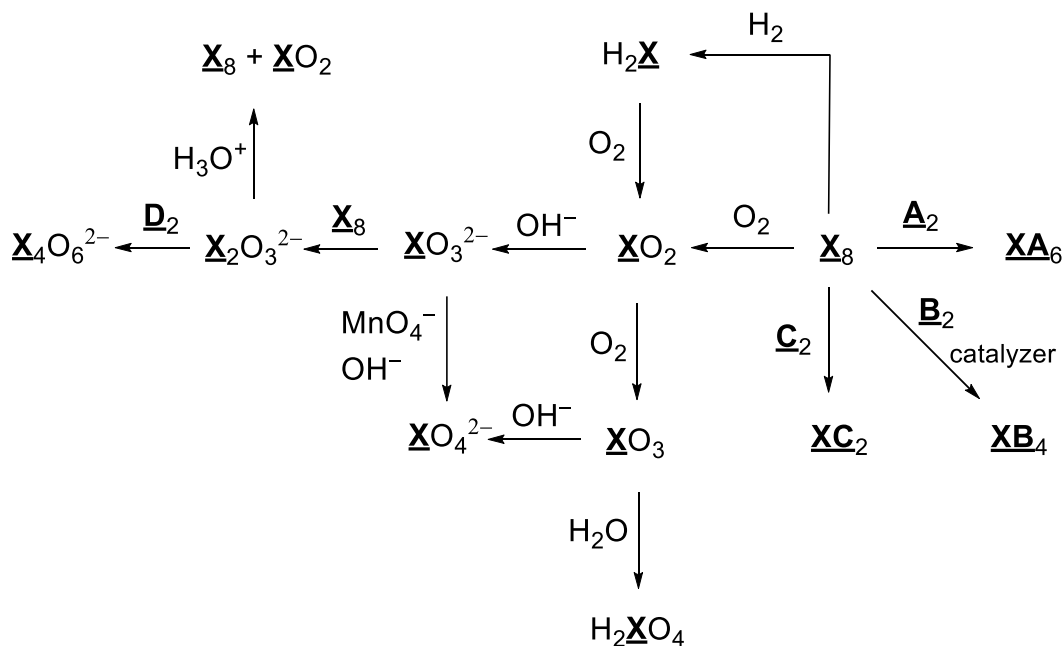
1.9. The only acidic hydroxide (non-amphoteric) formed by group 13 (IIIA) elements is H₃BO₃. **2 mark**

1.10. The binding energy of fluorine is relatively low/small among halogens even though the element fluorine has the smallest atomic radius of all the group 17 elements. **2 mark**

Question 2. “Here it smells to ...” (20% of the total; only 10th and 11th grades)

2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	2.10	Total	Total
10	10	4	24	4	6	2	4	6	10	80	20%
10	10	4	24	4	6	2	4	6	10	80	20%

The following diagram shows some of the chemical properties of the compounds of the chemical element X, whose simple substance X₈ is a yellow solid with a characteristic odor that has medicinal properties. The elements A, B, C and D are in the same group of the periodic table and the oxidizing power of their simple substances A₂, B₂, C₂ and D₂ decreases in that exact order.



2.1. Say the identity of the chemical elements represented by letters.

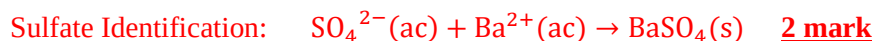
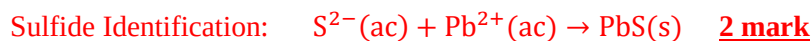
X = S 2 mark	A = F 2 mark	B = Cl 2 mark	C = Br 2 mark	D = I 2 mark
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2.2. Propose a mechanism for the reaction between the oxide XO₂ and C₂ in aqueous solution, forming the acids H₂XO₄ and HC. Formulate the formation of “logical” intermediaries.



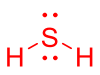
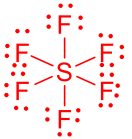
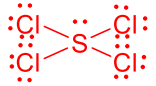
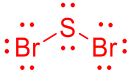
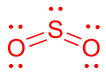
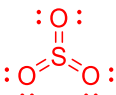
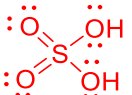
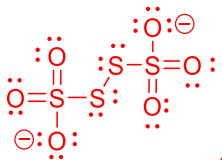
Other similar logical variants are accepted.

2.3. Create chemical test equations to identify species X²⁻ y XO₄²⁻ in the laboratory.



Other essays are accepted.

2.4. Represent the structures of the species H_2X , XA_6 , XB_4 , XC_2 , XO_2 , XO_3 , H_2XO_4 and $\text{X}_4\text{O}_6^{2-}$. For the first four, indicate the molecular geometry.

 angular <u>3 mark</u>	 octahedral <u>4 mark</u>	 irregular tetrahedral or seesaw <u>3 mark</u>	 angular <u>3 mark</u>	 <u>2 mark</u>
 <u>2 mark</u>	 <u>3 mark</u>	 <u>4 mark</u>		

2.5. Nitric acid in the presence of H_2XO_4 forms the nitronium cation NO_2^+ . Represent the resonant structures of this ion and classify it according to the Theory of Hard and Soft Acids and Bases (TABDB).

hard acid <u>1 mark</u>	$\left[:\ddot{\text{O}}=\text{N}=\ddot{\text{O}}: \longleftrightarrow :\text{O}\equiv\text{N}-\ddot{\text{O}}: \longleftrightarrow :\ddot{\text{O}}-\text{N}\equiv\text{O}: \right]^+$	<u>3 mark</u>
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2.6. State the adjusted ionic equation for the oxidation of XO_3^{2-} with permanganate in a basic medium.

$3\text{SO}_3^{2-}(\text{ac}) + 2\text{MnO}_4^-(\text{ac}) + \text{H}_2\text{O}(\text{l}) \rightarrow 3\text{SO}_4^{2-}(\text{ac}) + 2\text{MnO}_2(\text{s}) + 2\text{OH}^-(\text{ac})$	<u>6 mark</u>
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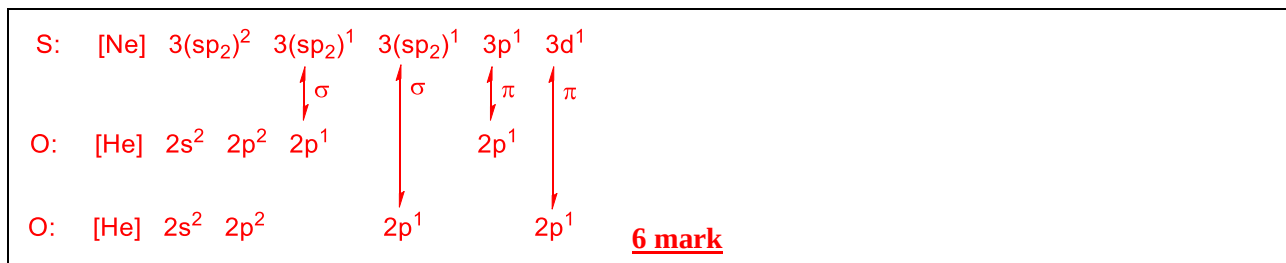
2.7. Compare the acid-base character of the oxides XO_2 and XO_3 based on the partial charges on the atoms in the molecules.

Sulfur(VI) is more electronegative than sulfur(IV), which causes less partial negative charge on oxygen (less basic) and greater partial positive charge on sulfur (more acidic). Therefore, trioxide is more acidic than dioxide. <u>2 mark</u>
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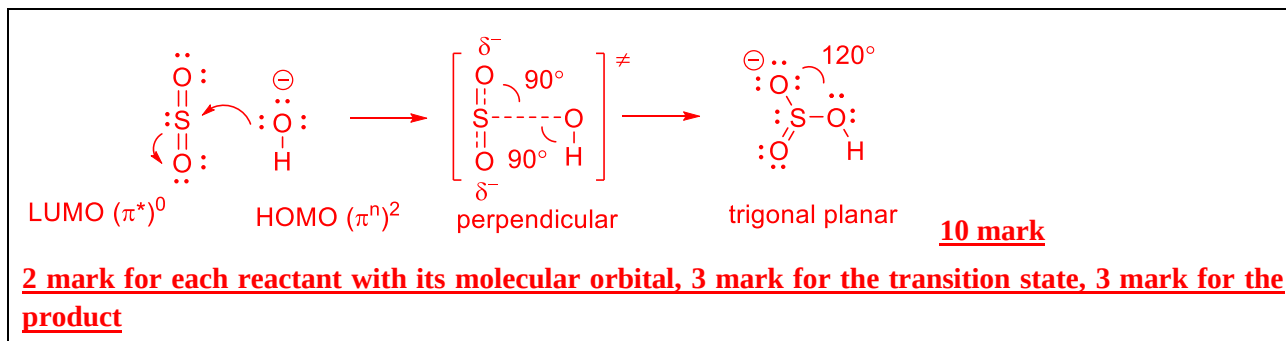
2.8. The anions $\text{X}_2\text{O}_3^{2-}$ and XO_4^{2-} are isoelectronic, however, the former reacts with I_2 and the latter does not react. Justify this fact according to the postulates of the TABDB.

Answer 1: Diiodine is a soft acid. Thiosulfate is an ambidentate ion in which an exposed sulfur atom can act as a soft basic center. On the other hand, sulfate is a hard base since only the oxygen atoms are exposed and can react. The most favored interaction will be between diiodine and thiosulfate. <u>4 mark</u>
Answer 2: Sulfate does not react with diiodine since it is a species of sulfur in its maximum oxidation state (VI) that cannot be oxidized, while in thiosulfate, the sulfur is not in the maximum oxidation state and can be oxidized with diiodine, which is reduced to iodide. <u>(solo 2 mark)</u>

2.9. Represent the electronic distribution of the $\underline{\text{X}}\text{O}_2$ molecule using the Valencia Bond Theory. **Take into account the structure you drew in section 2.4.** Use only hybrid orbitals to describe the electrons of the central atom.



2.10. $\underline{\text{X}}\text{O}_3^{2-}$ formation from $\underline{\text{X}}\text{O}_2$ in a basic medium occurs in two steps. First, the $\underline{\text{X}}\text{O}_2\cdot\text{OH}^-$ adduct is formed and then deprotonated by a hydroxide ion. Consider that the external electronic structure by molecular orbitals of $\underline{\text{X}}\text{O}_2$ is $(\dots)(\pi^b)^2(\pi^n)^4(\pi^n)^2(\pi^*)^0(\sigma^*)^0$ and that for ion OH^- is $(\sigma^n)^2(\sigma^b)^2(\pi^n)^4(\sigma^*)^0$. Represent the mechanism of the first reaction step. In your analysis, specify the frontier orbitals that interact for each reactant and classify them (HOMO/LUMO/SOMO), the direction of electron transfer, the geometry of the transition state, as well as the bonds that are broken and/or formed, and the geometry of the product adduct.



Question 3. The lattice energy of KBF₄ (10% of the total; 10th, 11th and 12th grades)

3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	Total	Total
4	4	2	8	2	2	2	10	6	40	10%
4	4	2	8	2	2	2	10	6	40	10%

The lattice energy or net energy (U°) is the energy involved in the formation or rupture of an ionic solid, taking as reference the isolated ions (in the gaseous state). The sign of U° depends on the convention used. From now on, consider that this quantity is negative and refers to the solid formation reaction. The lattice U° values can be estimated from Coulomb's law, assuming that the interaction between the ions that form the crystal is essentially electrostatic. The Born-Mayer equation contains specific empirical-theoretical parameters that depend on both the crystalline structure (spatial arrangement of the particles) of the solid and the electronics of the ions:

$$U^\circ = -\frac{N_A \cdot M \cdot |z_+| \cdot |z_-| \cdot e^2}{4 \cdot \pi \cdot \epsilon_0 \cdot r_0} \left(1 - \frac{\rho}{r_0}\right)$$

The constant ρ has a value of 35 pm for all alkali metal halides; $N_A = 6.0221 \cdot 10^{23}$ is Avogadro's number; M is Madelung's constant and is determined by the arrangement of the ions in the crystal (crystalline structure); z_+ , z_- are the respective charges of the cation and anion; r_0 is the internuclear distance between the two nearest oppositely charged ions; $\epsilon_0 = 8.854 \cdot 10^{-12}$ C/(m·V) is the electrical permittivity in vacuum, and e is the elementary charge of the electron, whose value is $1.602 \cdot 10^{-19}$ C.

3.1. Use the previous equation to show that the U° of potassium fluoride is close to -800 kJ/mol. Consider that M is 1.7476 and the interion distance in the lattice r_0 is 271 pm. 1 pm = 10^{-12} m

$$U^\circ(\text{KF}) = -\frac{6,0221 \cdot 10^{23} \cdot 1,7476 \cdot 1 \cdot 1 \cdot (1,602 \cdot 10^{-19})^2}{4 \cdot 3,1416 \cdot 8,854 \cdot 10^{-12} \cdot 271 \cdot 10^{-12}} \left(1 - \frac{35}{271}\right) = -780 \text{ kJ/mol} \quad \text{4 mark}$$

3.2. Usually, cations are smaller than anions, and the radii ratio is $r_+/r_- < 1$. This does not occur in the fluorides of the heavier metals. Calculate the value of r_+/r_- in potassium fluoride, knowing that the K^+ ion has an ionic radius (r_+) that is 5 pm greater than the ionic radius (r_-) of F^- .

$$r_+ + r_- = 271 \quad r_+ = r_- + 5 \quad \rightarrow \quad 2 \cdot r_- = 271 - 5 \quad \rightarrow \quad r_- = 133 \text{ pm} \quad r_+ = 138 \text{ pm}$$

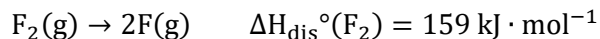
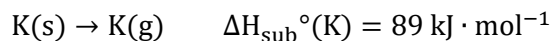
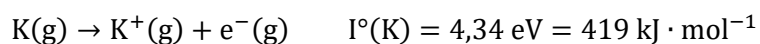
$$r_+/r_- = 1,04 \quad \text{4 mark}$$

3.3. There is a slight numerical difference between the lattice enthalpy ($\Delta H_{\text{U}}^\circ$) and U° since the latter is the variation of internal energy during the formation of the ionic solid. Although we will consider that both values are equal throughout this question, complete the following expression that relates both magnitudes for potassium fluoride by applying the First Principle of Thermodynamics.

$$\text{K}^+(\text{g}) + \text{F}^-(\text{g}) \rightarrow \text{KF}(\text{s}) \quad \Delta H_{\text{U}}^\circ(\text{KF}) = U^\circ(\text{KF})$$

$$\Delta H_{\text{U}}^\circ(\text{KF}) = U^\circ(\text{KF}) - 2 \cdot R \cdot T \quad \text{2 mark}$$

Thermochemical cycles allow us to find energy values that are difficult to measure. For example, the Born-Haber cycle relates U° to the enthalpy of formation of the ionic solid ($\Delta H_{\text{f}}^\circ$) and different atomic and molecular properties. Consider the following data:





We will use as a convention that the electronic affinity (A° , positive) is the opposite of the enthalpy of electronic gain ($\Delta H_{\text{ge}}^{\circ}$, negative).

3.4. Use the Born-Haber cycle and show that A° for the fluorine atom is close to 3.5 eV.

$$\Delta H_{\text{f}}^{\circ}(\text{KF}) = I(\text{K}) + \Delta H_{\text{sub}}^{\circ}(\text{K}) + 0,5 \cdot \Delta H_{\text{dis}}^{\circ}(\text{F}_2) - A(\text{F}) + U(\text{KF}) \quad \text{4 mark}$$

$$-567 = 419 + 89 + 0,5 \cdot 159 - A - 780 \quad \text{2 mark}$$

$$A = 374,5 \text{ kJ/mol} = 3,88 \text{ eV} \quad \text{2 mark}$$

According to Mulliken, the electronegativity (χ) equals the half-sum of the electron affinity and the ionization potential (in eV). For his part, Pearson established that hardness (η) is equal to the half-difference of the same properties to develop a mathematical basis for his Theory of Soft Hard Acids and Bases.

3.5. Calculate χ and η for the fluorine atom. $\chi(\text{F}) = \frac{I+A}{2}$ $\eta(\text{F}) = \frac{I-A}{2}$

The data of $I(\text{F}) = 17.42$ is not provided, so the question cannot be resolved. 2 mark

Using $I(\text{K})$ or other logical reasoning is accepted as a solution.

$$\chi(\text{F}) = \frac{I + A}{2} = \frac{17,42 + 3,88}{2} = 10,65 \quad \eta(\text{F}) = \frac{I - A}{2} = \frac{17,42 - 3,88}{2} = 6,77$$

3.6. Using your knowledge of chemical periodicity and the previous expressions, justify why fluorine is a harder species than chlorine.

$I(\text{F}) > I(\text{Cl})$ since the former attracts electrons more strongly. $A(\text{F}) < A(\text{Cl})$ product of the anomaly of the second period. Both factors contribute to making fluoride harder. 2 mark

The enthalpy of atomization ($\Delta H_{\text{atom}}^{\circ}$) can be defined as the heat evolved (at constant temperature and pressure) in breaking all the bonds of a molecule in the gaseous state to obtain the individual atoms that make up the molecule.



3.7. Assuming that the B-F bond is purely covalent, estimate the enthalpy of formation of the gaseous tetrafluoroborate ion ΔH_1° from the above data only.

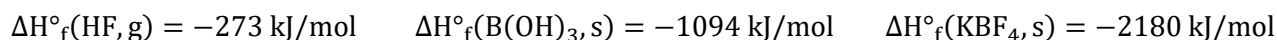
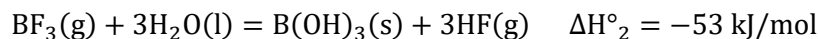


$$\Delta H_{\text{r}}^{\circ} = \Delta H_{\text{atom}}^{\circ}(\text{BF}_3) \approx 3 \cdot \Delta H^{\circ}(\text{B}-\text{F}) = 1938 \text{ kJ} \cdot \text{mol}^{-1} \rightarrow \Delta H^{\circ}(\text{B}-\text{F}) = 646 \text{ kJ} \cdot \text{mol}^{-1}$$

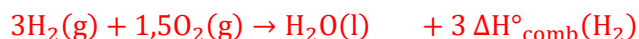


$$\Delta H_1^{\circ} \approx -\Delta H^{\circ}(\text{B}-\text{F}) = -646 \text{ kJ} \cdot \text{mol}^{-1} \quad \text{2 mark}$$

3.8. Calculate ΔH_3° for the formation of salt KBF_4 using Hess's Law and the following data.



A possible solution may be to decompose the problem reaction into simpler processes.



$$\Delta\text{H}_3^\circ = \Delta\text{H}_2^\circ + 3\Delta\text{H}_{\text{comb}}^\circ(\text{H}_2) - 3\Delta\text{H}_f^\circ(\text{HF}) - \Delta\text{H}_f^\circ(\text{boric acid}) + \Delta\text{H}_f^\circ(\text{KBF}_4) - \Delta\text{H}_f^\circ(\text{KF})$$

$$\Delta\text{H}_3^\circ = -53 + 3(-242) - 3(-273) - (-1094) + (-2180) - (-567) = -479 \text{ kJ/mol}$$

Note that the first terms are equivalent to the formation of BF_3 .

$$\Delta\text{H}_3^\circ = -\Delta\text{H}_f^\circ(\text{BF}_3) + \Delta\text{H}_f^\circ(\text{KBF}_4) - \Delta\text{H}_f^\circ(\text{KF})$$

8 mark for the solution route and 2 mark for the result.

3.9. Calculate U° for KBF_4 salt employing Hess' Law and the following data. If you could not answer items **3.1**, **3.7** and **3.8**, you may use for your calculations the values $U^\circ(\text{KF}) = -800 \text{ kJ/mol}$, $\Delta\text{H}_1^\circ = -650 \text{ kJ/mol}$ and $\Delta\text{H}_3^\circ = -450 \text{ kJ/mol}$.



$$U^\circ(\text{KBF}_4) = \Delta\text{H}_3^\circ + U^\circ(\text{KF}) - \Delta\text{H}_1^\circ \quad \textbf{4 mark}$$

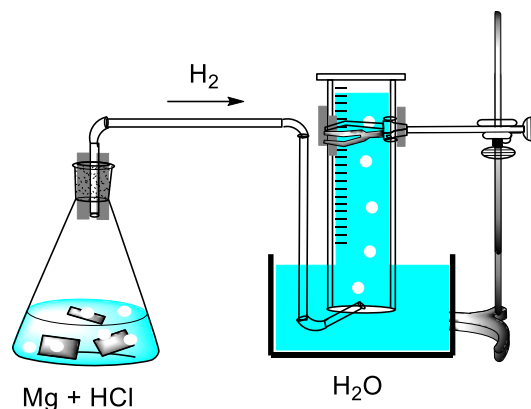
$$U^\circ(\text{KBF}_4) = -479 - 780 - (-646) = -613 \text{ kJ/mol} \quad \textbf{2 mark}$$

With the values offered, the result is -600 kJ/mol . Mixed results are accepted.

Question 4. Is it chemistry, mathematics or physics? (15% of total; 10th, 11th and 12th grades)

4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	4.10	Total	Total
2	2	8	4	10	4	10	4	4	12	60	15%
2	2	8	4	10	4	10	4	4	12	60	15%

In the chemistry laboratory of the National Training Center, a kinetic study of the reaction between magnesium and hydrochloric acid was carried out. For this purpose, the equipment shown in the figure was attached. To reduce errors, excess concentrated acid and fragments of different length l of the same tape (width $a = 3.5$ mm; thickness $h = 0.15$ mm) of metallic magnesium ($A_r = 24.3$; volumetric density $\rho = 1.74$ g/cm³ were used.). Next, the volume of dihydrogen formed as a function of time was determined for different experiments in which the variables temperature, acid concentration and exposed magnesium area were carefully controlled. The data obtained made it possible to calculate the initial reaction rates.



4.1. State the chemical equation of the reaction that occurs.



4.2. Why do magnesium strips left outdoors turn an ashy color?

Magnesium is oxidized by dioxygen in the air. 2 mark

4.3. Express the rate of the reaction v_R as a function of the variation of the volume of dihydrogen formed ΔV_{H_2} in a tiny time interval Δt , the temperature T , the atmospheric pressure p_T , the vapor pressure of water $p_{\text{H}_2\text{O}}$ and the volume of the acid solution V_D (consider this value as constant).

$$v_R = \frac{1}{2} \cdot \frac{dc(\text{HCl})}{dt} \approx \frac{1}{2} \cdot \frac{\Delta c(\text{HCl})}{\Delta t} \quad \text{para } \Delta t \approx 0$$

$$V(\text{H}_2) = \frac{n(\text{H}_2) \cdot R \cdot T}{p(\text{H}_2)} = \frac{0,5 \cdot n(\text{HCl}) \cdot R \cdot T}{p_{\text{tot}} - p(\text{H}_2)} \rightarrow \frac{\Delta V(\text{H}_2)}{\Delta t \cdot V_D} = \frac{0,5 \cdot \Delta n(\text{HCl}) \cdot R \cdot T}{[p_{\text{tot}} - p(\text{H}_2)] \cdot \Delta t \cdot V_D}$$

$$n(\text{HCl}) = 2 \cdot n(\text{H}_2) \rightarrow v_R = \frac{1}{2} \cdot \frac{\Delta c(\text{HCl})}{\Delta t} = \frac{\Delta V(\text{H}_2) \cdot [p_{\text{tot}} - p(\text{H}_2\text{O})]}{\Delta t \cdot V_D \cdot R \cdot T}$$

2 mark for each expression found (8 mark in total).

4.4. Calculate the surface densities σ (g/cm²) and linear densities λ (g/cm) of the magnesium tape used.

$$\rho = \frac{m}{V} = \frac{m}{l \cdot a \cdot h} \quad \lambda = \frac{m}{l} = \rho \cdot a \cdot h = 1,74 \cdot 0,35 \cdot 0,015 \approx 9,1 \cdot 10^{-3} \text{ g} \cdot \text{cm}^{-1} \quad \underline{\text{2 mark}}$$

$$\sigma = \frac{m}{A} = \rho \cdot h = 1,74 \cdot 0,015 \approx 0,026 \text{ g} \cdot \text{cm}^{-2} \quad \underline{\text{2 mark}}$$

By definition of surface density, you should only consider the area of one face in the calculation.

4.5. In some experiments, a piece of tape was used so that the total exposed magnesium surface was 70 mm². Calculate: i) the length l of the fragment, ii) the mass of magnesium it contains and iii) the volume (at 25 °C and 1.00 atm) of dihydrogen measured at the end of the reaction and the fragment being consumed.

$$A = 70 \text{ mm}^2 = 2(a \cdot h + a \cdot l + l \cdot h) = 2(3,5 \cdot 0,15 + 3,5 \cdot l + l \cdot 0,15) \rightarrow l = 9,45 \text{ mm} = 0,945 \text{ cm}$$

$$m(\text{Mg}) = \lambda \cdot l = 9,1 \cdot 10^{-3} \cdot 0,945 = 8,6 \cdot 10^{-3} \text{ g} = 3,54 \cdot 10^{-4} \text{ mol}$$

$$n(\text{H}_2) = n(\text{Mg}) = 3,54 \cdot 10^{-4} \text{ mol} \rightarrow V(\text{H}_2) = n(\text{H}_2) \cdot V_m = 3,54 \cdot 10^{-4} \cdot 22,5 = 0,0080 \text{ L}$$

--- Approximate calculation:

$$l = \frac{A}{2 \cdot a} = \frac{70 \text{ mm}^2}{2 \cdot 3,5 \text{ mm}} = 10 \text{ mm} \rightarrow m(\text{Mg}) = \lambda \cdot l = 9,1 \cdot 10^{-3} \cdot 1 = 0,0091 \text{ g} = 3,74 \cdot 10^{-4} \text{ mol}$$

$$n(\text{H}_2) = n(\text{Mg}) = 3,74 \cdot 10^{-4} \text{ mol} \rightarrow V(\text{H}_2) = n(\text{H}_2) \cdot V_m = 3,74 \cdot 10^{-4} \cdot 22,5 = 0,0084 \text{ L}$$

2 mark for length, 2 mark for mass, 2 mark for the amount of substance, 2 mark for the amount of dihydrogen, and 2 mark for volume.

4.6. Name all the laboratory supplies represented in the figure.

Rubber stopper, Erlenmeyer flask, hose or rubber section, extension clamp, crystallizer/beaker, universal support, nut or clamp. 4 mark (0.5 each) **4 mark (0,5 c/u)**

The results obtained in the experiments are summarized below:

Reaction	$[\text{H}_3\text{O}^+]$, mol/L	Surface area of Mg, mm^2	Temperature, $^\circ\text{C}$	Initial rate, mol/(L·s)
1	1,00	70	25,0	$1,56 \cdot 10^{-4}$
2	0,200	70	25,0	$5,64 \cdot 10^{-6}$
3	0,500	70	25,0	$3,36 \cdot 10^{-5}$
4	1,00	140	25,0	$2,79 \cdot 10^{-4}$
5	1,00	35	25,0	$7,30 \cdot 10^{-5}$
6	1,00	70	35,0	$2,67 \cdot 10^{-4}$
7	1,00	70	15,0	$1,19 \cdot 10^{-4}$

4.7. Determine the rate law and the constant k_R at 25°C . Round partial reaction orders to the nearest integer value. Say the units of k_R .

$$v_R = k_R \cdot A_{\text{Mg}}^{???} \cdot [\text{H}_3\text{O}^+]^{??}$$

$$\frac{\text{Exp 1}}{\text{Exp 2}} = \frac{1,56 \cdot 10^{-4}}{5,64 \cdot 10^{-6}} = \frac{k_R \cdot (70)^x \cdot (1,00)^y}{k_R \cdot (140)^x \cdot (0,2)^y} \rightarrow 27,6 = 5^y \rightarrow y \approx 2 \quad \text{2 mark}$$

$$\frac{\text{Exp 1}}{\text{Exp 4}} = \frac{1,56 \cdot 10^{-4}}{2,79 \cdot 10^{-4}} = \frac{k_R \cdot (70)^x \cdot (1,00)^y}{k_R \cdot (140)^x \cdot (1,00)^y} \rightarrow 0,559 = 0,5^x \rightarrow x \approx 1 \quad \text{2 mark}$$

$$v_R = k_R \cdot A_{\text{Mg}} \cdot [\text{H}_3\text{O}^+]^2 \quad \text{2 mark}$$

$$k_1 = 2,23 \cdot 10^{-6} \quad k_2 = 2,01 \cdot 10^{-6} \quad k_3 = 2,01 \cdot 10^{-6} \quad k_4 = 1,99 \cdot 10^{-6} \quad k_5 = 2,09 \cdot 10^{-6}$$

$$\bar{k}(25^\circ\text{C}) = 2,048 \cdot 10^{-6} \approx 2,0 \cdot 10^{-6} \text{ L} \cdot \text{mol}^{-1} \cdot \text{mm}^{-2} \cdot \text{s}^{-1}$$

2 mark for averaging, 1 mark for result and 1 mark for units (4 mark in total).

4.8. Assuming that the activation energy is 30 kJ/mol, how many times will the rate of the reaction increase when the temperature is increased from 25 to 35°C , keeping other variables constant?

$$\ln \frac{k'}{k} = -\frac{E_A}{R} \left(\frac{1}{T'} - \frac{1}{T} \right) = -\frac{30000}{8,3145} \left(\frac{1}{35 + 273,15} - \frac{1}{25 + 273,15} \right) = 0,393 \quad \text{2 mark}$$

$$\frac{k'}{k} = e^{0,393} \approx 1,5 \quad \text{2 mark}$$

4.9. Calculate the density (g/L) of dihydrogen at 25 °C and 1.00 atm pressure.

$$\rho(\text{H}_2) = \frac{M(\text{H}_2)}{V_m} = \frac{2,00}{24,5} = 0,0816 \text{ g} \cdot \text{L}^{-1}$$

2 mark for using the ideal gas equation or molar volume and 2 mark for the result (4 mark).

In their movement, the bubbles accelerate until they reach a maximum speed value when the forces are compensated. Opposing forces act on the bubbles. The thrust (E) favors the upward movement, and its module is numerically equal to the weight (F_g) of the volume of liquid displaced by the displaced bubble. Instead, the weight of the bubble and the friction force (F_d) oppose the movement. The friction force can be calculated using Stokes' Law, knowing that the dynamic viscosity coefficient of water is $\mu = 1.002 \cdot 10^{-3} \text{ Pa} \cdot \text{s}$.

$$\text{Fuerza de gravedad } F_g = m \cdot g$$

$$\text{Ley de Stokes } F_d = 6 \cdot \pi \cdot r \cdot \mu \cdot v$$

4.10. **Find an expression** and calculate the value of the maximum velocity $v_{\max}$ (in m/s) with which dihydrogen bubbles of radius $r = 5.0 \text{ mm}$ move inside the container. $\rho_{\text{H}_2\text{O}} = 1.00 \text{ g/cm}^3$, $g = 9.81 \text{ m/s}^2$

$$E = F_d + F_g(\text{gas})$$

$$F_g(\text{agua}) = F_d + F_g(\text{gas}) \quad \text{2 mark}$$

$$\rho_{\text{H}_2\text{O}} \cdot V_{\text{H}_2} \cdot g = 6 \cdot \pi \cdot r \cdot \mu \cdot v_{\max} + \rho_{\text{H}_2} \cdot V_{\text{H}_2} \cdot g \quad \text{2 mark}$$

$$\rho_{\text{H}_2\text{O}} \cdot \frac{4}{3} \cdot \pi \cdot r_{\text{H}_2}^3 \cdot g = 6 \cdot \pi \cdot r_{\text{H}_2} \cdot \mu \cdot v_{\max} + \rho_{\text{H}_2} \cdot \frac{4}{3} \cdot \pi \cdot r_{\text{H}_2}^3 \cdot g \quad \text{2 mark}$$

$$\rho_{\text{H}_2\text{O}} \cdot \frac{2}{3} \cdot r_{\text{H}_2}^2 \cdot g = 3 \cdot \mu \cdot v_{\max} + \rho_{\text{H}_2} \cdot \frac{2}{3} \cdot r_{\text{H}_2}^2 \cdot g \rightarrow v_{\max} = \frac{2}{9} \cdot \frac{(\rho_{\text{H}_2\text{O}} - \rho_{\text{H}_2}) \cdot r_{\text{H}_2}^2 \cdot g}{\mu} \quad \text{2 mark}$$

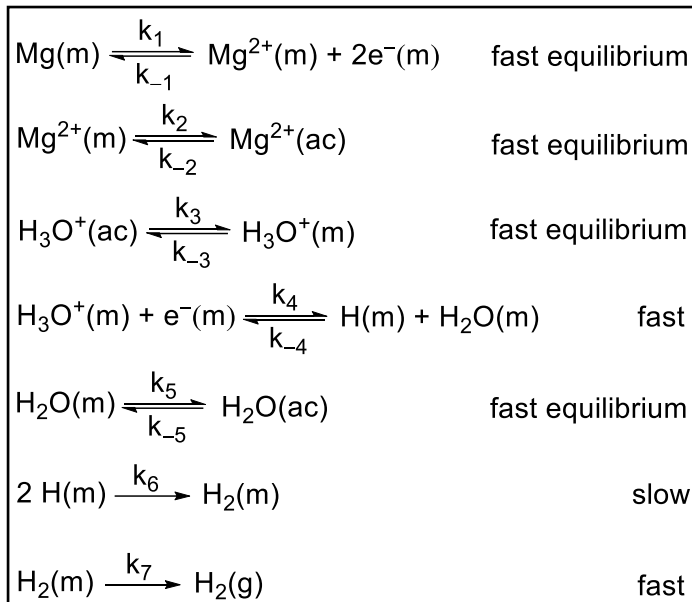
$$v_{\max} = \frac{2}{9} \cdot \frac{(1000 - 0,0816) \cdot (5,00 \cdot 10^{-3})^2 \cdot 9,81}{1,002 \cdot 10^{-3}} \approx 54 \text{ m/s}$$

1 mark for water density, 1 mark for bubble radius, 2 mark for result (4 mark).

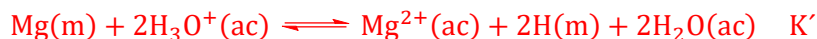
Question 4A. (5% of the total; only 12th grade)

4A.1	Total	Total
20	20	5%
20	20	5%

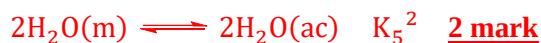
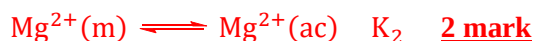
4A.1. For the previous reaction, the following mechanism has been proposed. The aggregation state of the species adsorbed on the metal surface is denoted as “(m)”. It is assumed that Mg^{2+} cations and free electrons exist on the surface of the metal. Consider that the ratio $[\text{Mg(m)}]/[\text{Mg}^{2+}(\text{ac})]$ is proportional to the area of reacted A_{Mg} metal. Use the pre-equilibrium approximation to express the approximate rate law in terms of the individual step constants. Treat the concentration of a species on the metal surface or in solution in an analogous manner.



A possible solution may be to calculate the equilibrium constant



Therefore, the reaction can be decomposed into several simpler processes.



$$K' = K_1 \cdot K_2 \cdot K_3^2 \cdot K_4^2 \cdot K_5^2 = \frac{[\text{Mg}^{2+}(\text{ac})] \cdot [\text{H}_2\text{O(ac)}]^2 \cdot [\text{H(m)}]^2}{[\text{Mg(m)}] \cdot [\text{H}_3\text{O}^{+}(\text{ac})]^2} \quad \underline{\text{2 mark}}$$

$$[\text{H(m)}]^2 = \frac{K'}{[\text{H}_2\text{O(ac)}]^2} \cdot \frac{[\text{Mg(m)}] \cdot [\text{H}_3\text{O}^{+}(\text{ac})]^2}{[\text{Mg}^{2+}(\text{ac})]} \quad \underline{\text{2 mark}}$$

$$v = k_6 \cdot [\text{H(m)}]^2 = k_6 \cdot \frac{K'}{[\text{H}_2\text{O(ac)}]^2} \cdot \frac{[\text{Mg(m)}] \cdot [\text{H}_3\text{O}^{+}(\text{ac})]^2}{[\text{Mg}^{2+}(\text{ac})]} \quad \underline{\text{2 mark}}$$

$$v = k_6 \cdot \frac{K'}{[\text{H}_2\text{O(ac)}]^2} \cdot k' \cdot A(\text{Mg}) \cdot [\text{H}_3\text{O}^{+}(\text{ac})]^2 \quad \underline{\text{2 mark}}$$

Where k' is a proportionality constant.

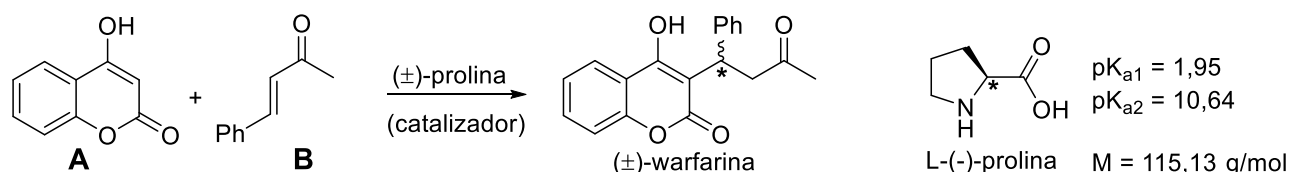
$$k_r = k_6 \cdot \frac{K'}{[\text{H}_2\text{O(ac)}]^2} \cdot k' \quad \underline{\text{2 mark}}$$

Question 5 Warfarin synthesis (20% of the total; only 12th grade)

5.1	5.2	5.3	5.4	5.5	5.6	5.7	5.8	5.9	5.10	5.11	5.12	5.13	5.14	5.15	5.16	Total	Total
6	2	2	2	6	8	2	6	8	6	2	4	4	2	16	4	80	20%
6	2	2	2	6	8	2	6	8	6	2	4	4	2	16	4	80	20%

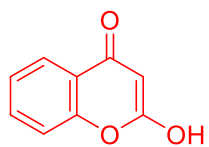
In the Joint Chemical and Biomolecular Synthesis Laboratory “José Luis Mola Garate” (SINTLAB) of the University of Havana (UH), the optimization of the synthesis process of active ingredients of several essential medicines is carried out to implement its production in Cuba and avoid costly imports. One of the substances under study is warfarin, a clotting inhibitor discovered by Karl Paul Link and his team in 1948 at the University of Wisconsin. Warfarin is derived from the natural product dicumarol, which is found in putrid sweet clover and causes death from excessive bleeding of animals that eat the plant. The exciting story of its discovery can be found in the BBC article “How the anticoagulant warfarin, the rat poison that doctors prescribe, was discovered.”

In the laboratory, the synthesis of warfarin is carried out by the reaction between 4-hydroxycoumarin (**A**) and trans-4-phenyl-3-buten-2-one (**B**), catalyzed by the natural amino acid proline.

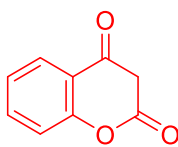


5.1. 4-Hydroxycoumarin can exist in the form of three tautomers. Why does the tautomer that appears in the diagram exist in a greater proportion? Represent the structures of the two remaining tautomers.

The tautomer represented is the most stable because it presents greater conjugation. **2 mark**



2 mark



2 mark

5.2. Name the reaction above and calculate the atomic economy (AE) percent.

Conjugate addition, 1,4 addition or Michael reaction. **1 mark**

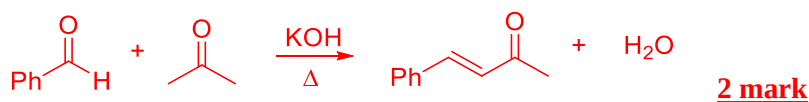
The atomic economy of this reaction is 100% since no side products are formed. **1 mark**

5.3. Why is warfarin obtained in racemate form in the above reaction?

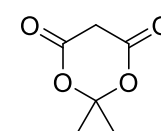
The reaction is carried out using achiral reactants and a catalyst. **2 mark**

It is acceptable to only refer to the catalyst in the answer.

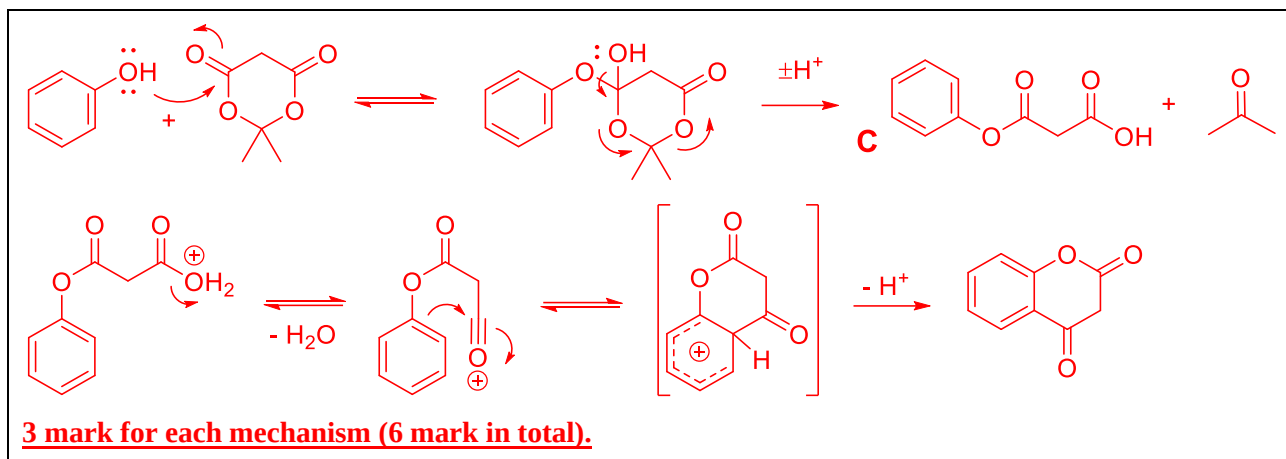
5.4. Write the adjusted equation for the condensation reaction that allows you to obtain reactant **B**.



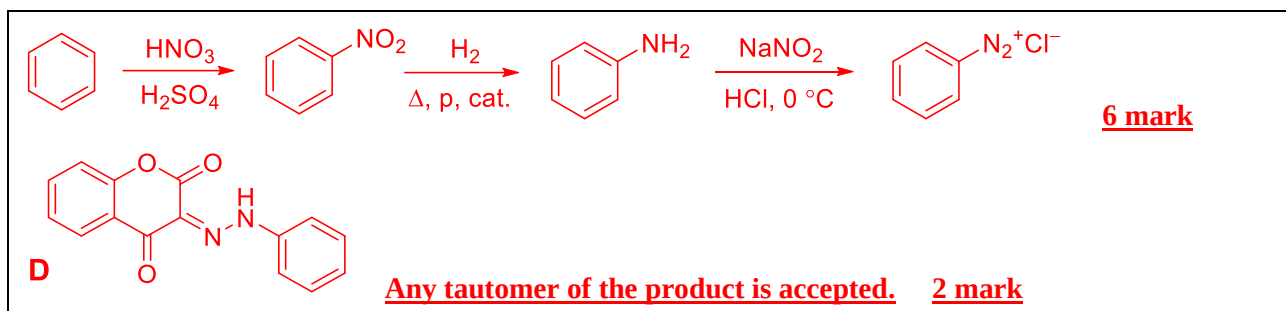
5.5. 4-Hydroxycoumarin can be obtained in two reaction steps by reacting phenol with Meldrum acid at high temperature, forming acetone and compound **C**, which undergoes an intramolecular Friedel-Crafts reaction in an acid medium. Propose a mechanism for both reactions.



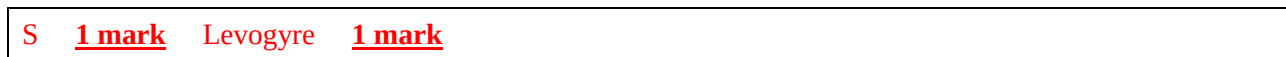
Meldrum' acid



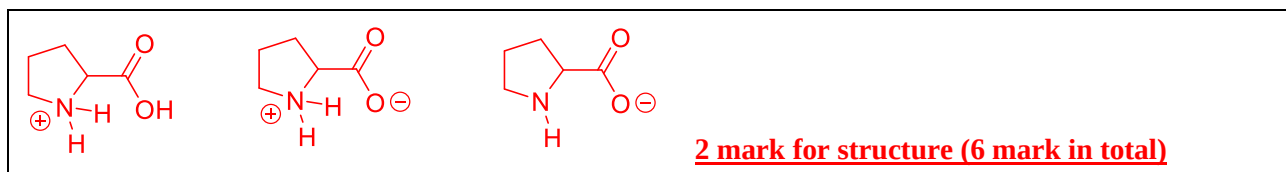
5.6. 4-Hydroxycoumarin gives rise to the diazotization reaction by the most nucleophilic carbon atom in its structure. Represent the synthesis of the simplest possible diazonium salt from benzene and the product **D** formed by its diazotization reaction with 4-hydroxycoumarin.



5.7. Proline in nature occurs mainly in the form of the L enantiomer, shown in the previous scheme. Classify the stereogenic center (*) according to its absolute configuration (R/S) and the proline according to the direction of the deviation of polarized light (right-handed/left-handed).

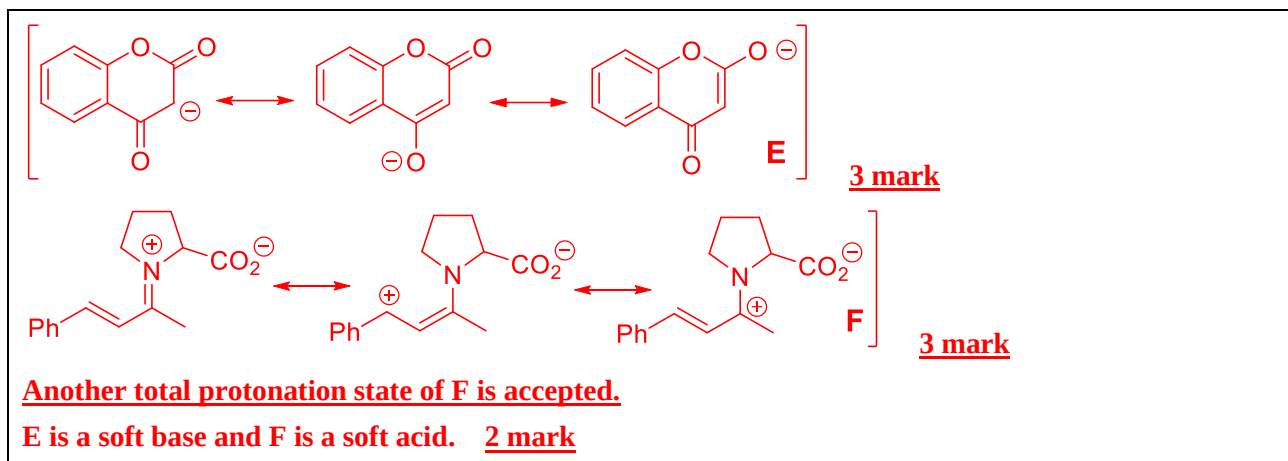


5.8. Proline exists in a neutral ionic form (zwitterion) in solid state and aqueous solution. In strongly acidic or basic aqueous media, proline is found in cationic or anionic form, respectively. Represent the three charged structures of proline described above.



In the main reaction of the synthesis, a relatively large amount (50 mol%) of proline is used, which performs the double function of base and organic catalyst. Proline deprotonates coumarin **A**, forming a very nucleophilic anion **E**. In contrast, as a product of the reaction between proline and the unsaturated ketone **B**, the formation of a very electrophilic intermediate **F** occurs.

5.9. Represent the resonant structures of the intermediates **E** and **F**. Classify them according to the Theory of Hard and Soft Acids and Bases.



5.10. Calculate the proportion in which the three forms of proline (cation, zwitterion, anion) are found in pure water (pH = 7.00) with respect to the total amount of proline dissolved (α_0 , α_1 , α_2 , respectively). Hint: Consider for your analysis that cationic proline is a diprotic acid.

$$\begin{aligned}
 &[H^+]^2 + K_{a1} \cdot [H^+] + K_{a1} \cdot K_{a2} \\
 &= [10^{-7}]^2 + 10^{-1,95} \cdot [10^{-7}] + 10^{-1,95} \cdot 10^{-10,64} = 10^{-14} + 10^{-8,95} + 10^{-12,59} \approx 10^{-8,95} \\
 &\chi(\text{cation}) = \frac{10^{-14}}{10^{-8,95}} = 8,9 \cdot 10^{-6} \quad \chi(\text{zwiter}) = \frac{10^{-8,95}}{10^{-8,95}} = 1,0 \quad \chi(\text{anión}) = \frac{10^{-12,59}}{10^{-8,95}} = 2,3 \cdot 10^{-4}
 \end{aligned}$$

2 mark for each fraction (6 mark in total).

5.11. The isoelectric point (pI) of a molecule is the pH value for which the predominant form is the neutral one, in the case of proline, the zwitterion. Calculate the isoelectric point of proline.

The zwitterionic species is the intermediate protonation state of proline; therefore

$$\text{pI} = \frac{\text{pK}_{a1} + \text{pK}_{a2}}{2} = \frac{1,95 + 10,64}{2} \approx 6,3 \quad \text{2 mark}$$

The partition or partition coefficient (K_D) is the ratio between the equilibrium concentrations of a substance X in the two phases of a mixture of two solvents (Y, Z) immiscible at equilibrium. It is often confused with the distribution coefficient (D), which is the ratio between the total concentrations of the substance, which includes all the ionized forms X_i in which it is found.

$$K_D = \frac{[X]_Y}{[X]_Z} \quad D = \frac{c(X)_{\text{total en Y}}}{c(X)_{\text{total en Z}}} = \frac{[X1]_Y + [X2]_Y + \dots}{[X1]_Z + [X2]_Z + \dots}$$

5.12. Estimate the partition coefficient of proline between water and dichloromethane if the solubility of proline is 162 g in 100 g of water ($\rho = 1.00$ g/ml) and 1.0 g in 100 g of dichloromethane (DCM, $\rho = 1.33$ g/ml).

Since the isoelectric point of proline is close to the pH = 7 value of pure water, the second acid dissociation constant is small and, using the result in part 5.10, all proline can be considered to be in zwitterion form. Assuming further that in dichloromethane, which has a low polarity, proline does not dissociate or form dimers, we have to

$$K_D = \frac{[\text{Pro}]_{\text{H}_2\text{O}}}{[\text{Pro}]_{\text{DCM}}} \approx D = \frac{c(\text{Pro})_{\text{total in H}_2\text{O}}}{c(\text{Pro})_{\text{total in DCM}}} = \frac{\frac{m(\text{Pro})}{M(\text{Pro}) \cdot V(\text{H}_2\text{O})}}{\frac{m(\text{Pro})}{M(\text{Pro}) \cdot V(\text{DCM})}} = \frac{\frac{S(\text{Pro})_{\text{H}_2\text{O}} \cdot \rho(\text{H}_2\text{O}) \cdot 10^3}{100}}{\frac{S(\text{Pro})_{\text{DCM}} \cdot \rho(\text{DCM}) \cdot 10^3}{100}}$$

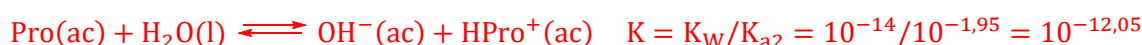
$$K_D = \frac{S(\text{Pro})_{\text{H}_2\text{O}} \cdot \rho(\text{H}_2\text{O})}{S(\text{Pro})_{\text{DCM}} \cdot \rho(\text{DCM})} = \frac{162 \cdot 1,00}{1,0 \cdot 1,33} = 122 \equiv K_D' = 0,00821 \text{ (inverse coefficient)}$$

Exact answer:

$$c(\text{Pro})_{\text{DCM}}^{\text{total}} = [\text{Pro}]_{\text{DCM}} = \frac{m(\text{Pro})}{M(\text{Pro}) \cdot V(\text{DCM})} = \frac{S(\text{Pro})_{\text{H}_2\text{O}} \cdot \rho(\text{DCM}) \cdot 10^3}{M(\text{Pro}) \cdot 100} = \frac{1,0 \cdot 1,33 \cdot 10^3}{115,13 \cdot 100}$$

$$[\text{Pro}]_{\text{DCM}} = 0,116 \text{ mol} \cdot \text{L}^{-1}$$

On the contrary, in water, the dissociation of proline occurs, favoring the formation of the anion.



However, both constants are very small and most of the proline does not dissociate; therefore

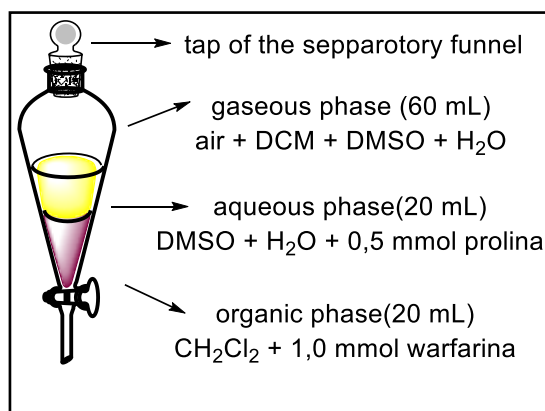
$$c(\text{Pro})_{\text{H}_2\text{O}}^{\text{total}} = [\text{Pro}]_{\text{H}_2\text{O}} = \frac{S(\text{Pro})_{\text{H}_2\text{O}} \cdot \rho(\text{H}_2\text{O}) \cdot 10^3}{M(\text{Pro}) \cdot 100} = \frac{162 \cdot 1,00 \cdot 10^3}{115,13 \cdot 100} = 14,1 \text{ mol} \cdot \text{L}^{-1}$$

$$K_D = \frac{[\text{Pro}]_{\text{H}_2\text{O}}}{[\text{Pro}]_{\text{DCM}}} = \frac{14,1}{0,116} = 122$$

5.13. Find an expression for the distribution coefficient of proline in the water/dichloromethane system based only on the constants K_{a1} , K_{a2} , pH and K_D .

$$D = \frac{c_{\text{tot}}(\text{Pro})_{\text{H}_2\text{O}}}{c_{\text{tot}}(\text{Pro})_{\text{DCM}}} = \frac{[\text{Pro}]_{\text{H}_2\text{O}}}{\alpha_1 \cdot [\text{Pro}]_{\text{DCM}}} = \frac{K_D \cdot (10^{-2 \cdot \text{pH}} + K_{a1} \cdot 10^{-\text{pH}} + K_{a1} \cdot K_{a2})}{K_{a1} \cdot 10^{-\text{pH}}} \quad \text{4 mark}$$

When warfarin synthesis is completed, the crude or mixture product of the reaction is poured onto water, and the resulting suspension is transferred to a separating funnel, where the product is extracted and washed. In one of the experiments at SINTLAB, a researcher used 1.0 mmol of each reactant and 5.0 ml of dimethyl sulfoxide (DMSO; $M = 78.13 \text{ g/mol}$; $\rho = 1.10 \text{ g/ml}$) as a solvent. Subsequently, he added 15.0 mL of water and transferred the resulting suspension to a separatory funnel with a capacity of 100 mL. Under these conditions, all the DMSO dissolves in the water and the product precipitates due to the change in the polarity of the medium. Then, the researcher added 20.0 ml of dichloromethane ($M = 84.93 \text{ g/mol}$; $\rho = 1.33 \text{ g/ml}$), closed the funnel



and shook vigorously to dissolve the product in the organic phase. However, the researcher was not careful, and due to the pressure difference between the gas inside the funnel and the outside air, the funnel tap was thrown off.

Fortunately, he did not lose any of the product and could collect the organic phase. Because the investigator wanted to recover as much warfarin as possible, he performed a second extraction with 20.0 mL of dichloromethane. This time, after shaking the mixture vigorously inside the separatory funnel, he carefully removed the tap and shook again, repeating these steps until the pressures inside and outside the container were equalized. He

then let it sit so that the organic and aqueous phases separated completely. Then, he combined the organic phases collected in the previous two extractions (40 ml), washed them with brine, and, after separating them in the funnel, dried the organic phase with anhydrous sodium sulfate. After evaporating all the solvent and leaving

the very high-purity solid obtained in a desiccator overnight, he calculated that the final yield of the process was 80%.

5.14. Why is warfarin more soluble in dichloromethane, while proline is more soluble in water?

Proline exists in a zwitterionic form and can form hydrogen bonds, making it very soluble in water. On the other hand, warfarin has nonpolar aromatic groups in its structure, so it dissolves better in a low polarity solvent such as dichloromethane. **4 mark**

5.15. Using the data shown in the figure, calculate the pressure of the gas phase inside the separating funnel after reaching equilibrium. Assume that before stirring, the gas phase contained only air (21 mol% dioxygen, 78 mol% dinitrogen, 1 mol% argon). The temperature during the experiments was 25.0°C and the air pressure inside the laboratory was 1.00 atm. The vapor pressure of DCM, water, and DMSO liquids is 46.7 kPa, 3.1664 kPa, and 0.06 kPa, respectively. Consider that Dalton's and Raoult's laws are valid, the volumes of liquids are additive, the ideal behavior of the gas phase and the dissolved solutes are non-electrolytes.

$$\text{Dalton's law} \quad p_{\text{total}} = \sum_i p_i^{\text{gas}} \quad \text{Raoult's law} \quad p_i^{\text{gas}} = x_i^{\text{liquido}} \cdot p_i^{\circ}$$

You must calculate: $p(\text{O}_2)$, $p(\text{N}_2)$, $p(\text{Ar})$, $p(\text{DCM})$, $p(\text{DMSO})$, $p(\text{H}_2\text{O})$ y $p(\text{total})$ in kPa.

--- In the aqueous phase (H_2O -DMSO-proline)

$$n(\text{H}_2\text{O}) = \frac{15,0 \text{ mL} \cdot 1,00 \text{ g} \cdot \text{mL}^{-1}}{18,0 \text{ g} \cdot \text{mol}^{-1}} = 0,8333 \text{ mol}$$

$$n(\text{DMSO}) = \frac{5,0 \text{ mL} \cdot 1,10 \text{ g} \cdot \text{mL}^{-1}}{78,13 \text{ g} \cdot \text{mol}^{-1}} = 0,0704 \text{ mol}$$

The amount of proline is 0.5 mmol (per data) and since it is not volatile, it does not exert pressure on the mixture.

$$n(\text{total}) = n(\text{H}_2\text{O}) + n(\text{Pro}) + n(\text{DMSO}) = 0,8333 + 0,0005 + 0,0704 = 0,9042 \text{ mol} \quad \textbf{3 mark}$$

$$\chi(\text{H}_2\text{O}) = \frac{0,8333}{0,942} = 0,922$$

$$p_{\text{H}_2\text{O}} = \chi(\text{H}_2\text{O}) \cdot P^{\circ}_{\text{H}_2\text{O}} = 0,922 \cdot 3,1664 = 2,92 \text{ kPa} \quad \textbf{2 mark}$$

$$\chi(\text{DMSO}) = \frac{0,0704}{0,9042} = 0,0779$$

$$p_{\text{DMSO}} = \chi(\text{DMSO}) \cdot P^{\circ}_{\text{DMSO}} = 0,0779 \cdot 0,06 = 0,00467 \text{ kPa} \quad \textbf{2 mark}$$

It is accepted to neglect the contribution of DMSO to the total pressure because it is very little volatile.

--- In the organic phase (DCM-warfarin)

The amount of warfarin is 1.0 mmol (per data) and since it is not volatile, it does not exert pressure on the mixture.

$$n(\text{DCM}) = \frac{20,0 \text{ mL} \cdot 1,33 \text{ g} \cdot \text{mL}^{-1}}{84,93 \text{ g} \cdot \text{mol}^{-1}} = 0,313 \text{ mol}$$

$$n(\text{total}) = n(\text{DCM}) + n(\text{warfarin}) = 0,313 + 0,001 = 0,314 \text{ mol} \quad \textbf{2 mark}$$

$$\chi(\text{DCM}) = \frac{0,313}{0,314} \approx 1,00 \rightarrow p_{\text{DCM}} \approx P^{\circ}_{\text{DCM}} = 46,7 \text{ kPa} \quad \textbf{2 mark}$$

--- The gas contribution from the air that initially filled the funnel will be

$$p_{\text{O}_2} = \chi(\text{O}_2) \cdot P_{\text{aire}} = 0,21 \cdot 101,325 = 21,278 \text{ kPa} \quad \textbf{1 mark}$$

$$p_{\text{Ar}} = \chi(\text{Ar}) \cdot P_{\text{air}} = 0,01 \cdot 101,325 = 1,013 \text{ kPa} \quad \underline{\text{1 mark}}$$

$$p_{\text{N}_2} = \chi(\text{N}_2) \cdot P_{\text{air}} = 0,78 \cdot 101,325 = 79,034 \text{ kPa} \quad \underline{\text{1 mark}}$$

--- The total pressure once liquid-vapor equilibrium is established is

$$p(\text{total}) = p_{\text{H}_2\text{O}} + p_{\text{DMSO}} + p_{\text{DCM}} + p_{\text{O}_2} + p_{\text{Ar}} + p_{\text{N}_2}$$

$$p(\text{total}) = 2,92 + 0,00467 + 46,7 + 101,325 = 151 \text{ kPa} \quad \underline{\text{2 mark}}$$

5.16. What is the maximum mass (in kg) that the tap of the funnel could have so that the pressure difference between the gas inside the container and the air outside would be able to trigger it? Consider that the tap is shaped like a cylinder with a diameter of 1.0 cm. Neglect the friction force between the tap and the upper part of the funnel. If you did not solve the previous part, assume that the pressure of the gas inside the funnel is $p(\text{total}) = 150 \text{ kPa}$.

$$F_g = E \rightarrow m(\text{tapa}) \cdot g = \Delta P \cdot A \quad \underline{\text{1 mark}}$$

$$\Delta P = P(\text{internal}) - P(\text{external}) = 151 - 101 = 50 \text{ kPa} \quad \underline{\text{1 mark}}$$

$$m(\text{tap}) = \frac{\Delta P \cdot A}{g} = \frac{\Delta P \cdot \pi \cdot r^2}{g} = \frac{50 \cdot 10^3 \cdot 3,1415 \cdot \left(\frac{0,01}{2}\right)^2}{9,81} \approx 0,40 \text{ kg} \quad \underline{\text{2 mark}}$$

Question 6. A cumbersome alloy... (10% of the total; only 10th grade)

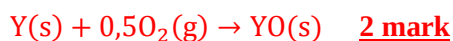
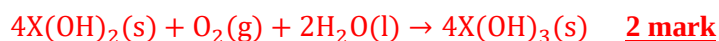
6.1	6.2	Total	Total
10	30	40	10%
10	30	40	10%

Stoichiometry emerged in the 18th century when the basic principles governing chemical processes were stated, including the law of conservation of mass, and laid the foundations of modern chemistry. In

the following problem, you must apply your knowledge of algebra and stoichiometry.

A 10.0 g sample of an alloy contains three metals of the first transition series (nickel, X and Y) in its composition. The sample is reacted with sufficient HCl, obtaining a solution that contains the chlorides of the metals Ni and X, with formulas NiCl_2 and XCl_2 , in addition to 0.0529 mol of H_2 . And it does not react with hydrochloric acid. NaOH is subsequently added to the solution, forming the reddish-brown hydroxide. On the other hand, when the remaining solid Y is heated in air, 8.701 g of the corresponding oxide of formula YO is formed.

6.1. Set up the **balanced** equations for the chemical reactions mentioned previously.



It is not necessary to pose the equation $\text{NiCl}_2\text{(ac)} + 2\text{NaOH(ac)} \rightarrow \text{Ni(OH)}_2\text{(s)} + 2\text{NaCl(ac)}$

6.2. Identify metals **X** and **Y** and calculate the alloy composition (mass %).

$$n(\text{O}_2) = \frac{P \cdot V(\text{O}_2)}{R \cdot T} = \frac{101,3 \cdot 0,110}{8,3145 \cdot 298,15} = 0,00450 \text{ mol} \quad \underline{2 \text{ mark}}$$

$$n(\text{X(OH)}_3) = 4 \cdot n(\text{O}_2) = 4 \cdot 0,0045 = 0,0180 \text{ mol} \quad \underline{2 \text{ mark}}$$

$$M(\text{X(OH)}_3) = \frac{m(\text{X(OH)}_3)}{n(\text{X(OH)}_3)} = \frac{1,913}{0,0180} = 106,27 \frac{\text{g}}{\text{mol}} \quad \underline{2 \text{ mark}}$$

$$A_r(\text{X}) = 106,36 - 3 \cdot A_r(\text{O}) - 3 \cdot A_r(\text{H}) = 106,36 - 3 \cdot 16,0 - 3 \cdot 1,0 = 55,27 \text{ g/mol} \quad \underline{2 \text{ mark}}$$

$$A_r(\text{Mn}) = 55,94 \text{ g/mol} \quad A_r(\text{Fe}) = 55,85 \text{ g/mol}$$

Given the data in the problem about reactivity, element **X** must be iron. $\underline{2 \text{ mark}}$

$$m(\text{Fe}) = A_r(\text{Fe}) \cdot n(\text{Fe}) = 55,85 \cdot 0,0180 = 1,01 \text{ g} \quad \underline{2 \text{ mark}}$$

$$n(\text{H}_2) = n(\text{Fe}) + n(\text{Ni}) \rightarrow n(\text{Ni}) = 0,0529 - 0,0180 = 0,0349 \text{ mol} \quad \underline{2 \text{ mark}}$$

$$m(\text{Ni}) = 0,0349 \cdot 58,69 = 2,05 \text{ g} \quad \underline{2 \text{ mark}}$$

$$m(\text{Y}) = 10,0 - m(\text{Fe}) - m(\text{Ni}) = 10 - 1,01 - 2,05 = 6,94 \text{ g} \quad \underline{2 \text{ mark}}$$

$$\omega(\text{Fe}) = 10,1\% \quad \underline{2 \text{ mark}} \quad \omega(\text{Ni}) = 20,5\% \quad \underline{2 \text{ mark}} \quad \omega(\text{Y}) = 69,4\% \quad \underline{2 \text{ mark}}$$

In the last chemical reaction, we have to

$$m(\text{YO}) - m(\text{Y}) = m(\text{O}) = 8,701 - 6,94 = 1,76 \text{ g} \quad \underline{2 \text{ mark}}$$

$$n(\text{YO}) = n(\text{Y}) = n(\text{O}) = \frac{m(\text{O})}{A_r(\text{O})} = \frac{1,76}{16,0} = 0,110 \text{ mol} \quad \underline{2 \text{ mark}}$$

$$A_r(\text{Y}) = \frac{6,94}{0,110} = 63,1 \frac{\text{g}}{\text{mol}} \rightarrow \text{Y} = \text{Cu} \quad A_r(\text{Cu}) = 63,55 \frac{\text{g}}{\text{mol}} \quad \underline{2 \text{ mark}}$$

Question 7. Camagüey es la ciudad... (10% of the total; only 11th grade)

7.1	7.2	7.3	Total	Total
2	12	26	40	10%
2	12	26	40	10%

The “tinajón” is the Camagüey symbol par excellence; it is the most rooted local representation, which is why Camagüey is known as the City of the Tinajones. The chemical composition of

a tinajón is interesting. Usually, the material contains three fundamental components and 2% impurities. When the Ibero-American contestants Lianet and Daniela began chemistry in their province in tenth grade, they became interested in the composition of a jar. They discovered that it contains an atomic solid with a covalent bond that is abundant in the Earth's crust.

7.1. Could you tell what solid it is?

Silicon dioxide **2 mark**

Daniela then inquired about kaolin, one of the significant components of the tinajones. Kaolin is a mixed oxide that has stoichiometric amounts (equal amount of substance) of aluminum and silicon, in addition to 55.78% (m/m) of oxygen and 1.56% (m/m) of hydrogen.

7.2. Calculate the empirical formula of kaolin, considering that its formula mass is 258.26 g/mol.

$$m(\text{Al}) + m(\text{Si}) = 100 - m(\text{O}) - m(\text{H}) = 100 - 55,78 - 1,56 = 42,66 \text{ g} \quad \mathbf{2 \text{ mark}}$$

$$26,98 \cdot n(\text{Al}) + 28,09 \cdot n(\text{Si}) = 42,66 \rightarrow n(\text{Al}) = n(\text{Si}) = \frac{42,66}{26,98 + 28,09} = 0,775 \text{ mol} \quad \mathbf{4 \text{ mark}}$$

$$n(\text{O}) = \frac{55,78}{16} = 3,49 \text{ mol} \quad \mathbf{1 \text{ mark}} \quad n(\text{H}) = \frac{1,56}{1,01} = 1,54 \text{ mol} \quad \mathbf{1 \text{ mark}}$$

$$n(\text{O}):n(\text{H}):n(\text{Al}):n(\text{Si}) = 4,5:2:1:1 = 9:4:2:2$$



For her part, Lianet was fascinated with micas, which can appear in different proportions in the same jar. The generic composition of some micas can be expressed by the generic formula $\underline{\text{X}}_2\underline{\text{Y}}_4\text{Z}_6\text{O}_{20}(\text{OH})_4$, where $\underline{\text{X}}$ is K or Na,

$\underline{\text{Y}}$ is Al, Mg or Fe and $\underline{\text{Z}}$ is Si or Al, fundamentally. In a workshop near Lianet's house, three micas are sold. Their elemental analysis is shown below.

Mica	Components (% in mass)
I	O (47,27); Si (27,67); Mg (14,95); K (9,62); H (0,49)
II	O (39,00); Fe (34,00); Al (21,93); OH (6,90); Na (4,67)
III	O (48,39); Al(6,81); Si (28,33); K (9,85); Mg (6,12); H (0,50)

7.3. Calculate the formulas of micas I, II and III. **Please note the generic composition given above.**

--- Assuming 100 g of mica I.

$$n(\text{O}) = \frac{47,27}{16,00} = 2,95 \text{ mol} \quad \mathbf{1 \text{ mark}} \quad n(\text{Si}) = \frac{27,67}{28,09} = 0,985 \text{ mol} \quad \mathbf{1 \text{ mark}}$$

$$n(\text{Mg}) = \frac{14,95}{24,31} = 0,615 \text{ mol} \quad \mathbf{1 \text{ mark}} \quad n(\text{K}) = \frac{9,62}{39,10} = 0,246 \text{ mol} \quad \mathbf{1 \text{ mark}}$$

$$n(\text{OH}^-) = n(\text{H}) = \frac{0,49}{1,01} = 0,485 \text{ mol} \quad \mathbf{1 \text{ mark}}$$

$n(\text{O}):n(\text{Si}):n(\text{Mg}):n(\text{K}):n(\text{H}) = 12:4:2,5:1:2 = 24:8:5:2:4$ $\text{K}_2\text{Mg}_5\text{Si}_8\text{O}_{20}(\text{OH})_4$ **3 mark**

--- Assuming 100 g of mica II.

$$n(\text{O}) = \frac{39,00}{16,00} = 2,44 \text{ mol} \quad \mathbf{1 \text{ mark}} \quad n(\text{Fe}) = \frac{34,00}{55,85} = 0,609 \text{ mol} \quad \mathbf{1 \text{ mark}}$$

$$n(\text{Al}) = \frac{21,93}{26,98} = 0,813 \text{ mol} \quad \mathbf{1 \text{ mark}} \quad n(\text{Na}) = \frac{4,67}{22,99} = 0,203 \text{ mol} \quad \mathbf{1 \text{ mark}}$$

$$n(\text{OH}^-) = n(\text{H}) = \frac{6,90}{17,01} = 0,406 \text{ mol} \quad \mathbf{1 \text{ mark}}$$

$n(\text{O}):n(\text{OH}^-):n(\text{Fe}):n(\text{Al}):n(\text{Na}) = 12:2:3:4:1 = 24:4:6:8:2$ $\text{Na}_2\text{Fe}_6\text{Al}_8\text{O}_{20}(\text{OH})_4$ **3 mark**

--- Suponiendo 100 g de la mica III.

$$n(\text{O}) = \frac{48,39}{16,00} = 3,02 \text{ mol} \quad \mathbf{1 \text{ mark}} \quad n(\text{Al}) = \frac{6,81}{26,98} = 0,252 \text{ mol} \quad \mathbf{1 \text{ mark}}$$

$$n(\text{Si}) = \frac{28,33}{28,09} = 1,01 \text{ mol} \quad \mathbf{1 \text{ mark}} \quad n(\text{K}) = \frac{9,85}{39,10} = 0,252 \text{ mol} \quad \mathbf{1 \text{ mark}}$$

$$n(\text{Mg}) = \frac{6,12}{24,31} = 0,252 \text{ mol} \quad \mathbf{1 \text{ mark}} \quad n(\text{OH}^-) = n(\text{H}) = \frac{0,50}{1,01} = 0,495 \text{ mol} \quad \mathbf{1 \text{ mark}}$$

$n(\text{O}):n(\text{Al}):n(\text{Si}):n(\text{K}):n(\text{Mg}):n(\text{H}) = 12:1:4:1:1:2 = 24:2:8:2:2:4$

$\text{K}_2\text{Mg}_2\text{Al}_2\text{Si}_8\text{O}_{20}(\text{OH})_4$ **4 mark**

Equivalent formulas are accepted.

A -1 mark is penalized if the generic composition provided is not considered.

Question 8. CSI in the chemical laboratory (10% of the total; only 10th grade)

8.1	8.2	8.3	Total	Total
10	20	10	40	10%
10	20	10	40	10%

Luis Manuel and Adriana are two students who were part of the 2023 Chemistry Pre-selection. On one occasion, they detected that the labels of five reagent bottles were damaged.

Not being able to read them and identify the substances in these bottles, they decided to consult the list of entry and exit of reagents from the warehouse. With this document, they determined that the substances were **barium carbonate**, **calcium sulfate**, **lead(II) acetate**, **potassium cyanide** and **sodium sulfide**. They immediately set about investigating what substance was in each bottle. To do this, they labeled each one with a different alphabet letter.

- Solids **A**, **C**, **D** and **E** were white, while solid **B** was pale yellow.
- **A**, **C** and **E** were odorless, while **B** smelled like rotten eggs and **D** smelled like bitter almonds.
- Compounds **A** and **C** were insoluble in water, while the others dissolved well.

At this point, the students were almost sure what substances **B**, **D**, and **E** were.

- Solid portions of **B**, **C** and **D** were dissolved in aqueous hydrochloric acid, forming a gas. In contrast, **A** was insoluble, **E** was only slightly soluble, and neither evolved a gas.

Then, the two students could identify each substance, although they performed additional tests.

- The addition of sulfuric acid to the aqueous solution of **E** formed a white precipitate. In comparison, the addition of sodium hydroxide formed a white precipitate, soluble in excess reagent.
- The mixture of aqueous solutions of **B** and **E** made it possible to obtain a black solid.
- A precipitate with a characteristic dark blue color was obtained by adding **D** to an aqueous solution of iron(II) and iron(III) chlorides.

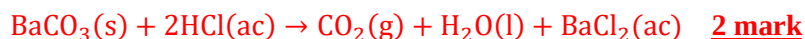
8.1. Identify the solids in jars **A**, **B**, **C**, **D**, and **E** by their chemical formula.

A = CaSO_4 **B** = Na_2S **C** = BaCO_3 **D** = KCN **E** = $\text{Pb}(\text{OAc})_2$

2 mark for each substance (10 mark in total).

A –0.5 mark is penalized for incorrect formulation.

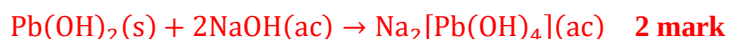
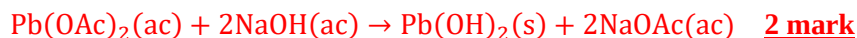
8.2. Formulate the adjusted equations of the chemical reactions carried out during the experiments.

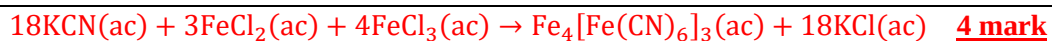


Note: This experiment can only be carried out under supervision and with appropriate means of protection due to the high toxicity of cyanide.



The smell of vinegar can also identify these reactions, which result from forming volatile acetic acid. However, this information is not necessary to solve the problem.





Other formulations of Prussian Blue are acceptable, for example, $\text{KFe}[\text{Fe}(\text{CN})_6]$ or $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$.

It is accepted to propose the ionic equations instead.

A -0.5 mark is penalized for poor fit, poor aggregation status or incorrect formulation.

8.3. In reality, the technicians could have identified each of the substances simply through the “flame test.” What colors characterize the cations of the previous salts?

Ba^{2+} : green yellow 2 mark

Ca^{2+} : Red brick 2 mark

Pb^{2+} : blue 2 mark

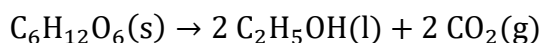
K^{+} : violet 2 mark

Na^{+} : yellow 2 mark

Question 9. Chemicals get along well with ethanol (12% of the total; only 10th grade)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9	9.10	Total	Total
4	2	5	6	3	8	6	4	2	8	48	12%
4	2	5	6	3	8	6	4	2	8	48	12%

Although biotechnology today is associated with the manufacture of vaccines and medicines, the use of living organisms to obtain compounds of interest is part of the early history of human civilization. One of the first compounds obtained was ethanol, used for recreational or medical purposes. Ethanol ($\text{CH}_3\text{-CH}_2\text{-OH}$, also formulated as $\text{C}_2\text{H}_5\text{OH}$) can be obtained from the fermentation of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) by different microorganisms:



Two students who participated in the last Central American and Caribbean Chemistry Olympiad, Diego and Royman, decided to synthesize ethanol. To do this, they took a 10.0 L container, filled it halfway with a glucose solution and placed it to ferment in a dark place at room temperature. Because they did not have the appropriate devices to allow the extraction of the gas formed during the reaction, one of their concerns was that the container had the proper strength for the process to be carried out.

9.1. Knowing that the container can withstand a pressure of 12 bar, calculate the maximum mass of ethanol ($M = 46.1 \text{ g/mol}$) that the students will produce at 25°C .

$$n(\text{CO}_2) = \frac{12 \cdot 100 \cdot 5}{8,3145 \cdot 298,15} = 2,42 \text{ mol} \quad \underline{\text{2 mark}}$$

$$n(\text{ethanol}) = n(\text{CO}_2) = 2,42 \text{ mol} \cdot 46,1 \text{ g} \cdot \text{mol}^{-1} = 111,6 \text{ g} \quad \underline{\text{2 mark}}$$

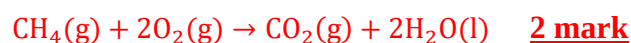
9.2. The ethanol obtained in this way is accompanied by various impurities. What separation method should students use to purify it? What physical property is it based on?

Distillation. 2 mark

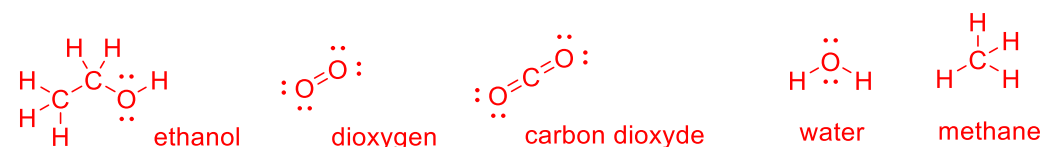
The method is based on the difference between boiling temperatures. 2 mark

Global ethanol production has increased because it is a renewable fuel to produce energy, and the depletion of fossil fuels such as coal (C), oil and natural gas, mainly composed of methane (CH_4).

9.3. State the equations for the complete combustion of 1 mole of ethanol, methane and carbon with dioxygen, forming carbon dioxide gas and liquid water.



9.4. Represent the structures of ethanol, methane, dioxygen, carbon dioxide, and water.



2 mark per ethanol structure and 1 mark for each other structure (6 mark in total).

Ethanol is miscible with water, meaning it can be mixed in any proportion and form a solution. On the contrary, dioxygen is slightly soluble in water, and carbon dioxide is somewhat more soluble than water.

9.5. Justify the different solubility in water of the substances ethanol, dioxygen and carbon dioxide based on the structures given in **9.4.** and intermolecular interactions.

Ethanol is a liquid substance made up of small, very polar molecules that form hydrogen bonds, so it is miscible with water. Carbon dioxide is a gaseous substance formed by nonpolar molecules that presents weak dipole-induced dipole interactions with water molecules. Still, it reacts as an acid, partially dissociating and is slightly soluble in water. Dioxygen is a gaseous substance formed by tiny nonpolar molecules and is only sparingly soluble in water. **3 mark**

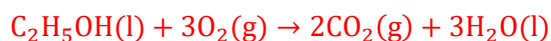
9.6. From the values of the average bond energies and the other thermodynamic quantities, estimate the values of the enthalpy change of the previous reactions in **9.3.**

Bond	O=O	C=O	C-O	C-C	C-H	O-H
E(A-B, kJ/mol)	495	799	355	348	414	463

$$\Delta H_{\text{atom}}^{\circ}(\text{C}) = 717 \text{ kJ/mol}, \Delta H_{\text{vap}}^{\circ}(\text{ethanol}) = 42,4 \text{ kJ/mol}, \Delta H_{\text{vap}}^{\circ}(\text{H}_2\text{O}) = 40,7 \text{ kJ/mol}$$

$$\Delta_r H^{\circ} \approx \sum E(\text{broken bonds}) - \sum E(\text{formed bonds}) \quad [\text{For reactions in gas phase}]$$

You must calculate $\Delta H_{\text{comb}}^{\circ}(\text{ethanol})$, $\Delta H_{\text{comb}}^{\circ}(\text{CH}_4)$ and $\Delta H_{\text{comb}}^{\circ}(\text{C})$ in kJ/mol.



$$\Delta_r H^{\circ} = \Delta_{\text{comb}} H^{\circ}(\text{ethanol}) \approx \Delta_{\text{vap}} H^{\circ}(\text{ethanol}) + 3 \cdot E(\text{O}=\text{O}) + E(\text{C}-\text{O}) + E(\text{O}-\text{H}) + \\ + E(\text{C}-\text{C}) + 5 \cdot E(\text{C}-\text{H}) - 2 \cdot 2 \cdot E(\text{C}=\text{O}) - 3 \cdot 2 \cdot E(\text{O}-\text{H}) - 3 \cdot \Delta_{\text{vap}} H^{\circ}(\text{water})$$

$$\Delta_r H^{\circ} = 42,4 + 3 \cdot 499 + 355 + 463 + 348 + 5 \cdot 414 - 4 \cdot 799 - 6 \cdot 463 - 3 \cdot 40,7$$

$$\Delta_r H^{\circ} = -1333 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{4 \text{ mark}}$$



$$\Delta_r H^{\circ} = \Delta_{\text{comb}} H^{\circ}(\text{C}) \approx \Delta_{\text{atom}} H^{\circ}(\text{C}) + E(\text{O}=\text{O}) - 2 \cdot E(\text{C}=\text{O})$$

$$\Delta_r H^{\circ} = 717 + 495 - 2 \cdot 799 = -386 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}}$$



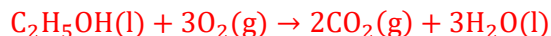
$$\Delta_r H^{\circ} = \Delta_{\text{comb}} H^{\circ}(\text{CH}_4) \approx 2 \cdot E(\text{O}=\text{O}) + 4 \cdot E(\text{C}-\text{H}) - 2 \cdot E(\text{C}=\text{O}) - 2 \cdot 2 \cdot E(\text{O}-\text{H}) \\ - 2 \cdot \Delta_{\text{vap}} H^{\circ}(\text{water})$$

$$\Delta_r H^{\circ} = 2 \cdot 495 + 4 \cdot 414 - 2 \cdot 799 - 4 \cdot 463 - 2 \cdot 40,7 = -885,4 \text{ kJ} \cdot \text{mol}^{-1} \quad \mathbf{2 \text{ mark}}$$

9.7. Calculate the previous values again (thermodynamically exact) using the enthalpies of formation corresponding to reactants and products.

$$\Delta_f H^{\circ} (\text{kJ/mol}): \text{ethanol} = -275; \text{CH}_4 = 79; \text{H}_2\text{O} = -286; \text{CO}_2 = -394.$$

You must calculate $\Delta H_{\text{comb}}^{\circ}(\text{ethanol})$, $\Delta H_{\text{comb}}^{\circ}(\text{CH}_4)$ and $\Delta H_{\text{comb}}^{\circ}(\text{C})$ in kJ/mol.



$$\Delta_r H^\circ = 2 \cdot \Delta_f H^\circ(\text{CO}_2) + 2 \cdot \Delta_f H^\circ(\text{H}_2\text{O}) - \Delta_f H^\circ(\text{ethanol})$$

$$\Delta_r H^\circ = 2 \cdot (-394) + 2 \cdot (-286) - (-275) = -1371 \text{ kJ} \cdot \text{mol}^{-1} \quad \underline{\text{2 mark}}$$



$$\Delta_r H^\circ = \Delta_f H^\circ(\text{CO}_2) = -394 \text{ kJ} \cdot \text{mol}^{-1} \quad \underline{\text{2 mark}}$$



$$\Delta_r H^\circ = \Delta_f H^\circ(\text{CO}_2) + 2 \cdot \Delta_f H^\circ(\text{H}_2\text{O}) - \Delta_f H^\circ(\text{methane})$$

$$\Delta_r H^\circ = (-394) + 2 \cdot (-286) - (-79) = -1045 \text{ kJ} \cdot \text{mol}^{-1} \quad \underline{\text{2 mark}}$$

9.8. Calculate the gravimetric (MJ/kg) and volumetric (MJ/L) energy densities of ethanol. Consider that $\Delta H_{\text{comb}}^\circ(\text{ethanol}) = -1400 \text{ kJ/mol}$ if you could not calculate it before and density 0.789 g/mL .

$$\frac{\Delta_{\text{comb}} H^\circ}{m(\text{ethanol})} = \frac{-(-1371) \text{ kJ} \cdot \text{mol}^{-1}}{46,1 \text{ g} \cdot \text{mol}^{-1}} \cdot 10^3 \text{ g} \cdot \text{kg}^{-1} = 2,97 \cdot 10^4 \frac{\text{kJ}}{\text{kg}} = 29,7 \frac{\text{MJ}}{\text{kg}} \quad \underline{\text{2 mark}}$$

$$\frac{\Delta_{\text{comb}} H^\circ}{V(\text{ethanol})} = 29,7 \frac{\text{MJ}}{\text{kg}} \cdot 0,789 \text{ kg} \cdot \text{L}^{-1} = 23,5 \frac{\text{MJ}}{\text{L}} \quad \underline{\text{2 mark}}$$

9.9. Given the gravimetric and volumetric energy densities of charcoal (26-33 MJ/kg, 34-43 MJ/L) and methane (55.6 MJ/kg, 0.0378 MJ/L; at 101.3 kPa and 15°C). Consider your calculated values for ethanol and decide which of the three fuels is most efficient.

Charcoal is the most efficient fuel from the point of view of energy density. However, it is a very polluting and non-renewable fuel. 2 mark

9.10. Royman and Diego decided to produce enough ethanol to heat all the water used in the first pre-selection stage for personal hygiene. Suppose that it is necessary to heat the water from 25°C to 40°C , that there are 150 people in the training center for 45 days and that each person consumes an average of 30 L of water. How much ethanol do they need? The heat capacity of water is $4.186 \text{ kJ}/(^\circ\text{C} \cdot \text{kg})$ and its density is 1.00 g/mL . Consider $\Delta H_{\text{comb}}^\circ(\text{C}_2\text{H}_6\text{O}) = -1400 \text{ kJ/mol}$ if you were unable to calculate it above.

$$Q_{\text{abs}} = -Q_{\text{ced}} \quad \underline{\text{2 mark}}$$

$$m(\text{H}_2\text{O}) \cdot \bar{C}(\text{H}_2\text{O}) \cdot \Delta T(\text{H}_2\text{O}) = V(\text{ethanol}) \cdot \left[\frac{\Delta_{\text{comb}} H^\circ}{V(\text{ethanol})} \right] \quad \underline{\text{2 mark}}$$

$$V(\text{ethanol}) = \frac{1,00 \frac{\text{kg}}{\text{L}} \cdot 150 \cdot 30 \cdot 45 \cdot 4,186 \frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}} \cdot (40 - 25)^\circ\text{C}}{23,5 \frac{\text{MJ}}{\text{L}} \cdot 10^3 \frac{\text{kJ}}{\text{MJ}}} = 541 \text{ L} = 427 \text{ kg} \quad \underline{\text{4 mark}}$$

A -1 mark will be penalized for each unit conversion error.

Question 10. Kinetics applied to enzymatic reactions (8% of the total; only 10th grade)

10.1	10.2	10.3	10.4	10.5	Total	Total
4	8	2	6	12	32	8%
4	8	2	6	12	32	8%

Enzymes are the rapid “mediators” of chemical reactions in our bodies. Enzymes bind to molecules (substrates) at specific

locations called catalytic sites and enable reactions to occur quickly. Studying these processes allows us to understand how our body metabolizes different substrates, which helps design disease treatments!

10.1. For a particular enzymatic reaction, it is known that its speed doubles with respect to the value of the speed at room temperature when the temperature increases by 10 °C. Considering that the system has Arrhenius-type behavior, calculate the activation energy (in kJ/mol).

$$E_A = \frac{-R \cdot \ln\left(\frac{k_2}{k_1}\right)}{\frac{1}{T_2} - \frac{1}{T_1}} = \frac{-8,3145 \cdot \ln 2}{\frac{1}{298,15 + 10} - \frac{1}{298,15}} = 52,9 \frac{\text{kJ}}{\text{mol}} \quad \mathbf{4 \text{ mark}}$$

10.2. Hydrogen peroxide is a very toxic metabolic product. The enzyme catalase catalyzes the breakdown of hydrogen peroxide into harmless water and dioxygen. Find the rate law and the value of the experimental specific rate constant from the initial rates and concentrations data in the table. **E** symbolizes the enzyme catalase.

v_0 ($\mu\text{M s}^{-1}$)	$[\text{H}_2\text{O}_2]_0$ (μM)	$[\text{E}]_0$ (μM)
$2,0 \cdot 10^7$	10,0	0,50
$8,0 \cdot 10^6$	5,0	0,20
$2,0 \cdot 10^7$	8,5	0,50
$1,2 \cdot 10^7$	10,0	0,30

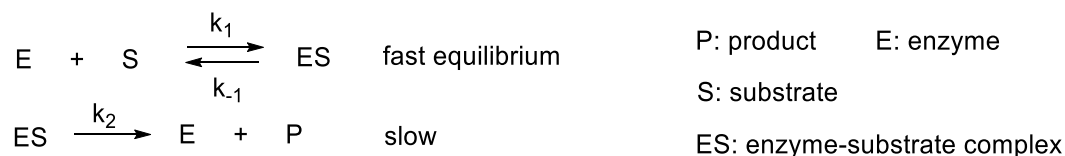
$$\frac{\text{Exp 1}}{\text{Exp 3}} = \frac{2 \cdot 10^7}{2 \cdot 10^7} = \frac{k}{k} \cdot \left(\frac{10}{8,5}\right)^a \cdot \left(\frac{0,5}{0,5}\right)^b \rightarrow a = 0 \quad \mathbf{2 \text{ mark}}$$

$$\frac{\text{Exp 1}}{\text{Exp 4}} = \frac{2 \cdot 10^7}{1,2 \cdot 10^7} = \frac{k}{k} \cdot \left(\frac{10}{10}\right)^0 \cdot \left(\frac{0,5}{0,3}\right)^b \rightarrow b = 1 \quad \mathbf{2 \text{ mark}}$$

$$v = k \cdot [\text{E}] \quad \mathbf{2 \text{ mark}}$$

$$k = 4,0 \cdot 10^7 \text{ s}^{-1} \quad \mathbf{2 \text{ mark}}$$

A mathematical model called “Michaelis-Menten kinetics” describes the kinetic behavior of enzymatic reactions. In 1913, Michaelis and Menten assumed an equilibrium between the enzyme, the substrate and the enzyme-substrate complex, formed instantaneously and maintained over time. The dissociation of the enzyme-substrate complex into enzyme and product is too slow to alter this balance. The model is summarized below:



The relationship $K_M = \frac{k_{-1} + k_2}{k_1}$ is called the Michaelis constant and in some cases, it numerically represents the inverse of the affinity of the substrate for the enzyme.

1. Assuming rapid chemical equilibrium: $k_1[\text{E}][\text{S}] = k_{-1}[\text{ES}]$

2. Separating constants from variables:

$$\frac{[\text{E}][\text{S}]}{[\text{ES}]} = \frac{k_{-1}}{k_1} = K_S \quad \text{where } K_S \text{ is the dissociation constant of the ES complex}$$

3. Substituting $[E] = [E]_0 - [ES]$, What is the mass balance equation and where $[E]_0$ is the initial (total) concentration of the enzyme

$$\frac{([E]_0 - [ES])[S]}{[ES]} = K_S$$

4. Solving for the concentration of the $[ES]$ complex, which is very difficult to measure experimentally:

$$[ES] = \frac{[E]_0[S]}{K_S + [S]}$$

5. Since the dissociation step is the slow one, we have $v = k_2[ES]$ and therefore:

$$v = \frac{k_2[E]_0[S]}{K_S + [S]} \quad (1)$$

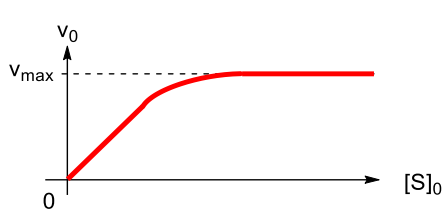
10.3. Why can it be assumed that $K_S = K_M$ under the conditions assumed by the model?

$k_2 \ll k_{-1}$ because the first step is a fast equilibrium and the second step is slow. **2 mark**

Suppose the initial substrate concentration $[S]_0$ is assumed to be much greater than the initial enzyme concentration $[E]_0$. In that case, the formation of the enzyme-substrate complex will result in a negligible change in the free substrate concentration. Therefore, we can replace $[S] = [S]_0$ in the previous expression. Taking into account that initial velocities are often determined, equation (1) becomes:

$$v_0 = \frac{k_2[E]_0[S]_0}{K_S + [S]_0} \quad (2)$$

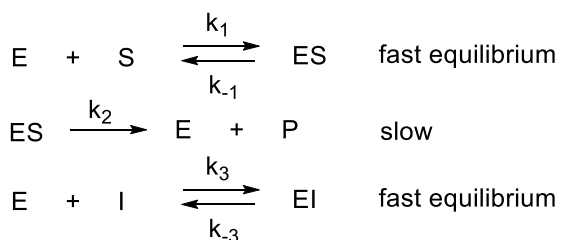
10.4. From equation (2), find an expression that describes the reaction rate v_0 i) at very low substrate concentrations and ii) at very high substrate concentrations (maximum rate, $v_{\max}$). Plot the approximate shape of a v_0 versus $[S]_0$ graph.

low concentration $v_0 = \frac{k_2}{K_S} \cdot [E]_0[S]_0$ 2 mark	high concentration $v_0 = k_2 \cdot [E]_0$ 2 mark	plot  2 mark
--	--	---

A competitive inhibitor is a substance that combines with the free enzyme to form a new enzyme-inhibitor EI complex and blocks the binding of the enzyme to the substrate. Under these conditions, the mass balance equation takes the form:

$$[E]_0 = [E] + [ES] + [EI]$$

and the reactions that occur are the following:



P: product E: enzyme

S: substrate I: inhibitor

ES: enzyme-substrate complex

EI: ES: enzyme-inhibitor complex

The constants K_S and K_M are defined as described above, while the new dissociation constant of the enzyme-inhibitor complex is defined as: $\frac{[E][I]}{[EI]} = \frac{k_{-3}}{k_3} = K_I$

10.5. Find an expression analogous to equation (1) for the rate law in a system where the enzyme, the substrate, and a competitive inhibitor are present. Assume the substrate concentration is much greater than the enzyme and inhibitor concentrations.

$$[E]_0 = [E] + [ES] + [EI] \quad \text{2 mark}$$

$$[E]_0 = [E] + \frac{[E][S]}{K_S} + \frac{[E][I]}{K_I} = [E] \left(1 + \frac{[S]}{K_S} + \frac{[I]}{K_I} \right) \quad \text{4 mark (2 mark per each substitution)}$$

$$v = k_2[ES] = k_2 \frac{[E][S]}{K_S} \quad \text{2 mark}$$

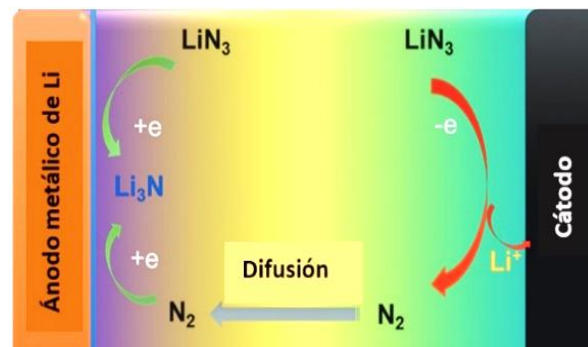
$$v = k_2 \frac{[E]_0[S]}{K_S \left(1 + \frac{[S]}{K_S} + \frac{[I]}{K_I} \right)} = k_2 \frac{[E]_0[S]}{K_S + \frac{K_S[I]}{K_I} + [S]} \quad \text{2 mark}$$

$$v = \frac{k_2[E]_0[S]}{K_S \left(1 + \frac{[I]}{K_I} \right) + [S]} \quad \text{2 mark}$$

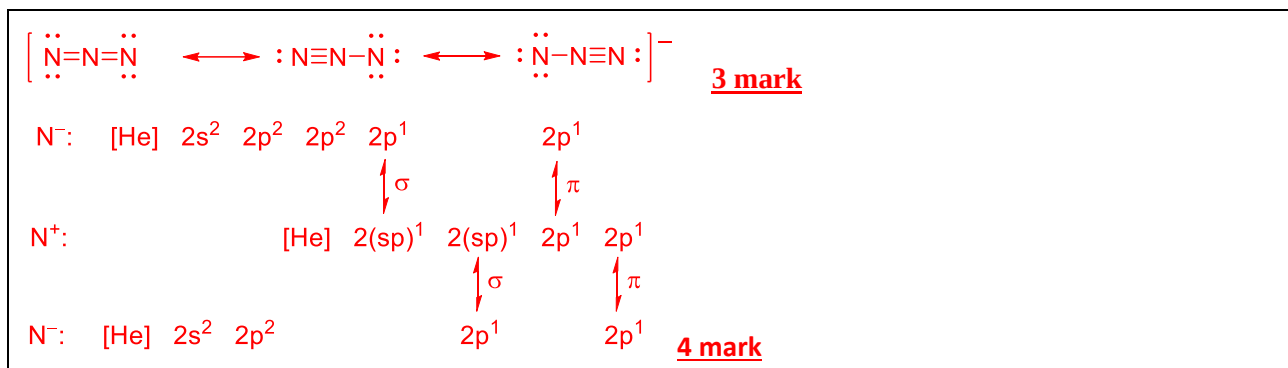
Question 11. Lithium azide (8% of total; only 11th grade)

11.1	11.2	11.3	11.4	11.5	Total	Total
7	7	7	2	9	32	8%
7	7	7	2	9	32	8%

Many researchers in the field of electrochemistry consider that lithium-sulfur batteries may be a new generation of rechargeable lithium batteries due to their high energy density. However, one of the fundamental problems in its design is the stability of the $\text{Li}^{(0)}$ metal anode. Lithium azide LiN_3 ($M = 48.97 \text{ g/mol}$) has been proposed as a possible electrolyte additive in solid-state Li-S batteries. This compound forms a passive layer of Li_3N nitride, which is thin, compact, and has excellent conductivity and covers the lithium anode. The coating of the electrode prevents its deterioration due to the formation of dendrites and the action of polysulfides, increasing the efficiency of the battery and its useful life. The figure schematically illustrates the operation of this type of battery. (Angew. Chem. Int. Ed. 2017, 56(48), 15368-15372; DOI: 10.1002/anie.201709305).



11.1. Represent the Lewis structure and the resonant structures of the azide ion. Make the electronic distribution (of nl^x type) according to the Valencia Bond Theory using hybrid orbitals for the central atom. Consider the formal charges of atoms.



For the design of a battery, a dilute aqueous solution of lithium azide was prepared at 2.0% (m/V) from a more concentrated initial solution of molality 14.0 mol/kg and density 0.707 g/cm^3 at room temperature (25°C).

11.2. Calculate the volume (in μL) of the initial solution required to prepare 125 μL of the 2% solution. Consider that 1% (m/V) = 1 g / 100 mL. $1 \mu\text{L} = 10^{-6} \text{ L}$, $1 \text{ mL} = 1 \text{ cm}^3$

$$m(\text{LiN}_3) = \frac{2 \text{ g}}{100 \text{ mL}} \cdot 125 \mu\text{L} \cdot 10^{-3} \frac{\text{mL}}{\mu\text{L}} = 2,5 \cdot 10^{-3} \text{ g} \quad \text{1 mark}$$

$$\omega_{\text{conc}}(\text{LiN}_3) = \frac{b \cdot M(\text{LiN}_3)}{1000 + b \cdot M(\text{LiN}_3)} = \frac{14 \cdot 48,97}{1000 + 14 \cdot 48,97} = 0,407 \quad \text{2 mark}$$

$$\rho_{\text{conc}}(\text{LiN}_3) = \omega_{\text{conc}}(\text{LiN}_3) \cdot d_{\text{conc}}(D) = 0,407 \cdot 0,707 = 0,288 \text{ g/mL} \quad \text{2 mark}$$

$$V_{\text{conc}}(D) = \frac{m(\text{LiN}_3)}{\rho_{\text{conc}}(\text{LiN}_3)} = \frac{2,5 \cdot 10^{-3} \text{ g}}{0,288 \text{ g/mL}} = 8,69 \cdot 10^{-3} \text{ mL} \approx 8,7 \mu\text{L} \quad \text{2 mark}$$

The solubility curve of lithium azide in water is shown below. Use the data provided in the graph to answer the following questions.

11.3. Classify the initial 14 mol/kg solution according to the degree of saturation. Justify numerically.

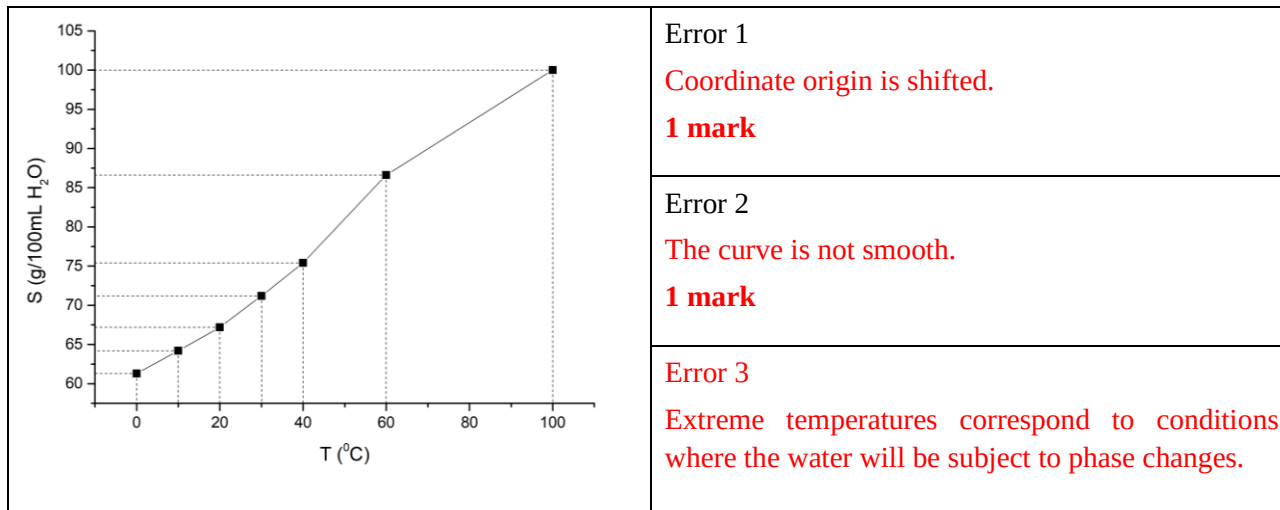
According to the solubility curve, at 25°C the solubility coefficient is 70g/100g water. **2 mark**

$$b(\text{LiN}_3) = \frac{n(\text{LiN}_3)}{m(\text{agua}) \cdot 10^{-3}} = \frac{m(\text{LiN}_3)}{m(\text{agua}) \cdot M(\text{LiN}_3) \cdot 10^{-3}} = \frac{S(\text{LiN}_3)}{100 \cdot M(\text{LiN}_3) \cdot 10^{-3}} \quad \mathbf{2 \text{ mark}}$$

$$b(\text{LiN}_3) = \frac{70}{100 \cdot 48,97 \cdot 10^{-3}} = 14 \frac{\text{mol}}{\text{kg}} \quad \mathbf{2 \text{ mark}}$$

The solution is approximately saturated. **1 mark**

11.4. Name two errors made during the construction of the graph.



11.5. 100 mL of the initial 14 mol/kg solution was heated until approximately half of the solvent present had evaporated. It was then cooled to 0°C and filtered under reduced pressure. What amount of solid lithium azide could recover after the procedure?

$$m(D) = V(D) \cdot d(D) = 100 \text{ mL} \cdot 0,707 \text{ g/mL} = 70,7 \text{ g} \quad \mathbf{1 \text{ mark}}$$

$$m(\text{water}) = [1 - \omega(\text{LiN}_3)] \cdot m(D) = [1 - 0,407] \cdot 70,7 = 41,9 \text{ g} \quad \mathbf{1 \text{ mark}}$$

$$m_{\text{total}}(\text{LiN}_3) = m(D) - m(\text{water}) = 70,7 - 41,9 = 28,8 \text{ g} \quad \mathbf{1 \text{ mark}}$$

After concentrating the solution, the mass of water is reduced to 21,0 g. **1 mark**

According to the solubility curve, at 25°C the solubility coefficient is 63g/100g water. **2 mark**

$$\frac{63}{100} = \frac{m_{\text{dis}}(\text{LiN}_3)}{m(\text{water})} = \frac{28,8 - m_{\text{pp}}(\text{LiN}_3)}{20,8} \quad \mathbf{2 \text{ mark}}$$

$$m_{\text{pp}}(\text{LiN}_3) = 15,7 \text{ g that precipitate.} \quad \mathbf{1 \text{ mark}}$$

Question 12. First transition series (12% of the total; only 11th grade)

12.1	12.2	12.3	12.4	12.5	12.6	12.7	12.8	12.9	12.10	12.11	12.12	Total	Total
5	2	6	2	2	2	4	3	4	10	6	2	48	12%
5	2	6	2	2	2	4	3	4	10	6	2	48	12%

12.1. Zinc in an acid medium is an excellent reducing agent, capable of reducing the oxidation state of the ions of the first transition series in aqueous solution. Fill in the colors of the ionic species formed during these processes. (In the table, the symbol X means instability).

Estado	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
VII	---	---	---	---	purple	---	---	---	---	---
VI	---	---	---	orange	green	X	---	---	---	---
V	---	---	yellow	---	X	---	---	---	---	---
IV	---	colorless	blue	---	brown	---	---	---	---	---
III	colorless	purple	green	green	X	yellow	X	X	X	---
II	---	X	purple	blue	pink	green	pink	green	blue	colorless
I	---	---	---	---	---	---	---	---	red	---

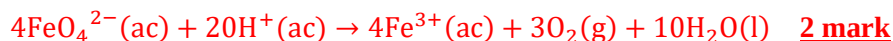
0.5 mark for each color (5 mark in total).

12.2. Throughout the series, the oxidizing power of the species increases. Because?

The increase in the effective nuclear charge of the metal causes the species to attract electrons more strongly and be better oxidants. **2 mark**

12.3. The ferrate(VI), cobalt(III) and nickel(III) ions are unstable in an aqueous solution and cause the evolution of gas **A**. On the other hand, titanium(II) ions are also unstable in aqueous solution but cause the evolution of gas **B**. Identify both gases and propose a chemical equation that exemplifies the formation of each one.

A = O₂ **1 mark** Any of the following equations are accepted, adjusted in an acidic or basic medium



B = H₂ **1 mark** Any of the following equations are accepted, adjusted in an acidic or basic medium



12.4. Why do ions in oxidation states IV-VII exist as oxygenated cations/anions?

Because the cation exerts a high polarizing effect on the water of hydration, causing the formation of strong covalent bonds. / Because the chemical bond in high oxidation states is fundamentally covalent due to the increase in electronegativity that occurs. / Because highly charged cations are very hard species that interact very strongly with the oxide ion. **2 mark**

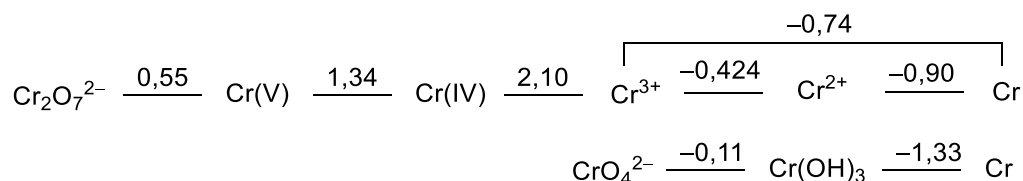
12.5. Titanium(IV) chloride is a liquid at room temperature, while titanium chlorides in lower oxidation states are solids. Explain this difference between the aggregation states.

Titanium(IV) chloride is a molecular substance whose particles have weak Van der Waals interactions and, therefore, have low melting and boiling temperatures. On the other hand, titanium(II) and (III) chlorides are ionic substances between whose particles there are strong electrostatic interactions and, therefore, have high melting and boiling temperatures. **2 mark**

12.6. Why are the oxides and hydroxides of the above chemical elements generally basic, amphoteric or acidic when the oxidation states are low, intermediate or high, respectively?

As the oxidation state increases, the electronegativity of the element increases. Consequently, the strength of the metal-oxygen bond increases, there is a decrease in the partial negative charge of oxygen and an increase in the partial positive charge of hydrogen. These factors favor acidic dissociation to the detriment of basic dissociation. **2 mark**

Latimer diagrams allow you to obtain relevant information. Use the diagrams of chromium in acidic and basic medium (E° in V, $T = 25^\circ\text{C}$) to answer the following questions.



12.7. Which chromium species are unstable in aqueous solution?

$E^\circ(\text{O}_2/\text{H}_2\text{O}) = 1,23 \text{ V (pH = 0); } 0,40 \text{ V (pH = 14)}$

$E^\circ(\text{H}_2\text{O}/\text{H}_2) = 0,00 \text{ V (pH = 0); } -0,82 \text{ V (pH = 14)}$

Cr(V) , Cr(IV) and Cr(0) are unstable in acidic environments. **3 mark**

Cr(0) is unstable in basic media. **1 mark**

12.8. Show that the standard potential for the redox couple $\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}$ is 1,33 V.

$$E^\circ(\text{Cr}^{\text{VI/III}}) = \frac{E^\circ(\text{Cr}^{\text{VI/V}}) + E^\circ(\text{Cr}^{\text{V/IV}}) + E^\circ(\text{Cr}^{\text{IV/III}})}{3} = \frac{0,55 + 1,34 + 2,1}{3} = 1,33 \text{ V} \quad \textbf{3 mark}$$

12.9. Show that the solubility product constant of chromium(III) hydroxide is $1,2 \cdot 10^{-30}$.



$$E = E^\circ(\text{Cr}^{\text{III/0}}) - \frac{0,0592}{3} \cdot \log\left(\frac{1}{[\text{Cr}^{3+}]}\right)$$

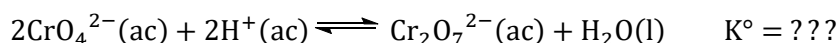
In a basic medium $[\text{OH}^-] = 1 \text{ mol/L}$ and it follows that

$$E = E^\circ(\text{Cr}^{\text{III/0}}) - \frac{0,0592}{3} \log\left(\frac{[\text{OH}^-]^3}{K_{\text{ps}}}\right) \quad \textbf{2 mark}$$

$$-1,33 = -0,74 - \frac{0,0592}{3} \log\left(\frac{1,00^3}{K_{\text{ps}}}\right) \rightarrow K_{\text{ps}} = 1,2 \cdot 10^{-30} \quad \textbf{2 mark}$$

Another solution may be to use a thermodynamic cycle.

12.10. Knowing that the ionic product of water is $K_{\text{W}} = 1,00 \cdot 10^{-14}$ at 25°C , calculate the formation constant of the dichromate ion from the condensation of two chromate ions in an acid medium:



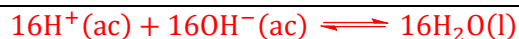
$$K_1^\circ = \left[10^{\frac{3 \cdot (-0,11)}{0,0592}}\right]^2 = 10^{-11,15} \quad \textbf{2 mark}$$



$$K_2^\circ = \left[10^{\frac{6 \cdot 1,33}{0,0592}}\right]^{-1} = 10^{-134,80} \quad \textbf{2 mark}$$



$$K_3^\circ = [1,2 \cdot 10^{-30}]^2 = 1,44 \cdot 10^{-60} \quad \textbf{2 mark}$$



$$K_4^\circ = [1,0 \cdot 10^{-14}]^{-16} = 10^{224} \quad \underline{\text{2 mark}}$$

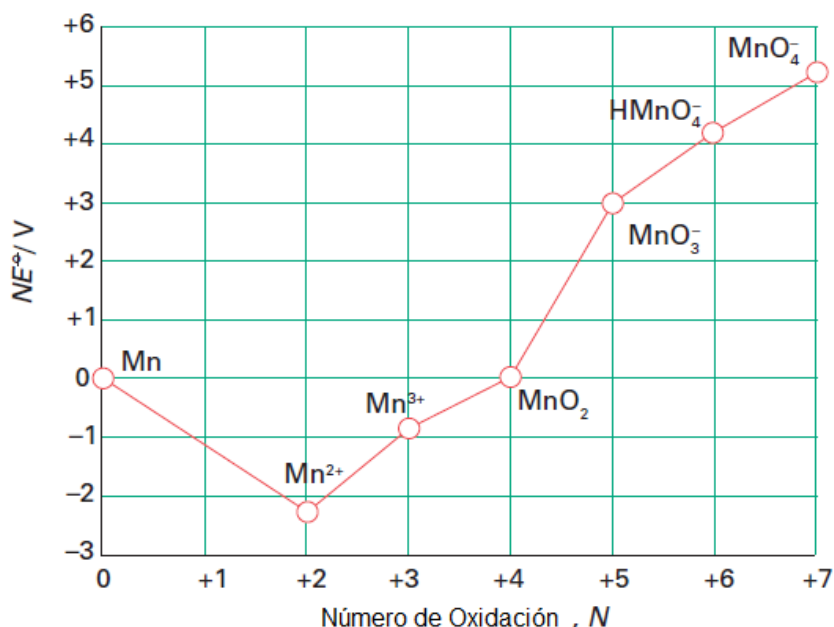


$$K^\circ = \prod K_i^\circ = 10^{-11,15} \cdot 10^{-134,80} \cdot 1,44 \cdot 10^{-60} \cdot 10^{224} = 1,44 \cdot 10^{18,05} \approx 1,6 \cdot 10^{18} \quad \underline{\text{2 mark}}$$

Another solution may be to use a thermodynamic cycle.

The Frost diagrams of an element. This type of representation is handy in determining which species are good oxidizing or reducing agents, in addition to providing the possibility of quickly identifying the thermodynamic stability of a given species under certain conditions.

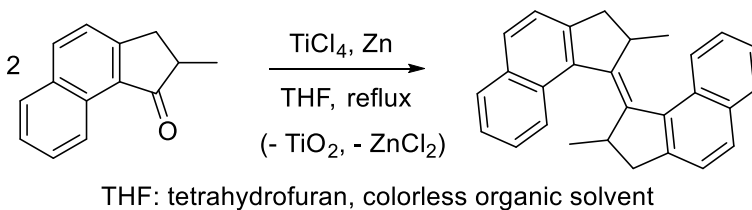
Use the Frost diagram of manganese referenced to the H^+/H_2 redox couple in an acidic solution ($\text{pH} = 0$) to answer the following questions.



12.11. Which species of manganese meet the following characteristics?

Stronger oxidant: MnO_3^- <u>0,5 mark</u>	Stronger reductant: Mn <u>0,5 mark</u>
More thermodynamically stable in a dihydrogen atmosphere: Mn^{2+} <u>0,5 mark</u>	
More thermodynamically stable in a dioxygen atmosphere: it cannot be determined. <u>0,5 mark</u>	
Unstable with respect to disproportion: Mn^{3+} , MnO_3^- and HMnO_4^- <u>1,5 mark</u>	
They combine to form MnO_2 : Mn^{2+} and MnO_4^- <u>1 mark</u>	
They combine to form Mn^{2+} : Mn and MnO_2 <u>1 mark</u>	
Insoluble in acidic aqueous solution: MnO_2 <u>0,5 mark</u>	

The above metals are widely used in organic reactions. For example, the McMurry reaction uses a salt of titanium and zinc to form alkenes. Nobel Laureate Ben L. Feringa used this reaction to synthesize first-generation molecular motors (J. Am. Chem. Soc. 2003, 125(49), 15076-15086; DOI: 10.1021/ja036782o). Initially, the solution of the reactants was colorless until the zinc powder was added and the reaction began to occur.



12.12. About McMurry's reaction, please respond.

- | |
|--|
| i) What is the function of zinc? Reductant. <u>1 mark</u> |
| ii) Why are Ti(IV) species colorless, while Ti(II) and Ti(III) species are colored?
Titanium(IV) has no electrons in the 3d orbitals that can be excited by visible light. <u>1 mark</u> |

Question 13. Water in the Swiss Alps (15% of the total; 11th and 12th grades)

13.1	13.2	13.3	13.4	13.5	13.6	13.7	13.8	13.9	13.10	Total	Total
5	10	4	3	6	6	8	10	6	2	60	15%
5	10	4	3	6	6	8	10	6	2	60	15%

When Erik and César went to the International Chemistry Olympiad last July, they discovered glaciers on Mount Säntis, more than two kilometers above sea level. To their surprise, the ice remained intact during the mentors' visit to the mountain, even though the temperature was slightly above 0 °C. Intrigued by this discovery, they decided to dust off Peter Atkins' old Physical Chemistry book that was abandoned on the bookshelf and scrutinize its pages in search of answers.

First, they found that atmospheric pressure decreases with increasing height h with respect to sea level according to the barometric equation. In the case of the troposphere, where the temperature varies approximately linearly with a ratio $L_b = 6.5 \text{ K} / 1000 \text{ m}$.

Variation of pressure with height in the troposphere

$$P' = P_b \cdot \left[\frac{T_b - h \cdot L_b}{T_b} \right]^{\frac{g \cdot M}{R \cdot L_b}}$$

where P is the pressure at height h , above sea level; $P_b = 101.325 \text{ kPa}$ and $T_b = 288.15 \text{ K}$ are the reference pressure and temperature, respectively; $g = 9.80665 \text{ m/s}^2$ is the acceleration of Earth's gravity; $M = 0.0289644 \text{ kg/mol}$ is the average molecular mass of air; and R is the gas constant.

13.1. Calculate the height (in m) of Mount Säntis if the barometric pressure P' at the summit is 74.7 kPa.

$$74,7 = 101,325 \cdot \left[\frac{288,15 - h \cdot \frac{6,5}{1000}}{288,15} \right]^{\frac{9,80665 \cdot 0,0289644}{8,314 \cdot \frac{6,5}{1000}}} = 5,256 \quad \text{1 mark}$$

$$\ln 74,7 = \ln 101,325 + 5,256 \cdot \ln \left(\frac{288,15 - h \cdot \frac{6,5}{1000}}{288,15} \right) \quad \text{1 mark}$$

$$e^{-0,05800} = 0,9436 = \frac{288,15 - h \cdot \frac{6,5}{1000}}{288,15} \quad \text{1 mark}$$

$$h \approx 2500 \text{ m} \quad \text{2 mark}$$

Then, Erik came up with an equation that allows us to calculate the phase transition temperature when the pressure differs from the standard pressure.

Clausius-Clapeyron' equation

$$\frac{dP}{dT} = \frac{\Delta S_{\text{trans}}}{\Delta V_{\text{trans}}}$$

where dP/dT represents the slope of the pressure vs temperature curve when two phases coexist in equilibrium; ΔS_{trans} and ΔV_{trans} are the entropy and volume changes for the phase transition, respectively.

For cases of change of state of aggregation of a pure substance, considering that ΔS_{trans} and ΔV_{trans} are approximately constant in a small temperature range and if the vapor in equilibrium behaves like an ideal gas, the previous equation is transformed to:

$$\text{Change of solid-liquid state: } \frac{P - P^\circ}{T_{\text{fus}} - T^\circ_{\text{fus}}} = \frac{\Delta H^\circ_{\text{fus}}}{\Delta V^\circ_{\text{fus}}} \quad \text{Change of liquid-vapor state: } \ln\left(\frac{P}{P^\circ}\right) = -\frac{\Delta H^\circ_{\text{vap}}}{R} \cdot \left(\frac{1}{T_{\text{eb}}} - \frac{1}{T^\circ_{\text{eb}}}\right)$$

where $T^\circ_{\text{fus}} = 273,15 \text{ K}$ and $T^\circ_{\text{eb}} = 373,15 \text{ K}$ are the melting and boiling temperatures at standard pressure $P^\circ = 1,00 \text{ bar} = 100 \text{ kPa}$, respectively; $\Delta H^\circ_{\text{fus}}$ and $\Delta H^\circ_{\text{vap}}$ are the standard enthalpies of fusion and vaporization; $\Delta V^\circ_{\text{fus}}$ is the volume change that occurs when fusion occurs.

	$\Delta H^\circ_f / \text{kJ} \cdot \text{mol}^{-1}$	$S^\circ / \text{J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	$\rho / \text{g} \cdot \text{dm}^{-3}$
$\text{H}_2\text{O(s)}$	-291,8 (25 °C)	43,2 (0 °C)	917
$\text{H}_2\text{O(l)}$	-285,8 (25 °C)	65,2 (0 °C) 69,9 (25 °C) 86,8 (100 °C)	1000
$\text{H}_2\text{O(g)}$	-241,8 (25 °C)	196,9 (100 °C)	0,598

13.2. Calculate $\Delta H^\circ_{\text{fus}}$, $\Delta S^\circ_{\text{fus}}$ (in $\text{J}/(\text{mol} \cdot \text{K})$), $\Delta V^\circ_{\text{fus}}$ (in dm^3/mol), $\Delta H^\circ_{\text{vap}}$ and $\Delta S^\circ_{\text{vap}}$ (in $\text{J}/(\text{mol} \cdot \text{K})$) for water from the thermodynamic data above.



$$\Delta H^\circ_{\text{fus}}(\text{H}_2\text{O}) = -285,8 - (-291,8) = 6,0 \text{ kJ} \cdot \text{mol}^{-1} \quad \text{2 mark}$$

$$\Delta S^\circ_{\text{fus}}(\text{H}_2\text{O}) = \frac{\Delta H^\circ_{\text{fus}}(\text{H}_2\text{O})}{T_{\text{fus}}} = \frac{6000}{273,15} = 22 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \quad \text{2 mark}$$

$\Delta S^\circ_{\text{fus}}$ can also be calculated by subtracting the absolute entropies of liquid and solid at 0 °C.

$$\Delta V^\circ_{\text{fus}}(\text{H}_2\text{O}) = M(\text{H}_2\text{O}) \cdot \left[\frac{1}{\rho_{\text{liq}}} - \frac{1}{\rho_{\text{sol}}} \right] = 18,02 \cdot \left[\frac{1}{917} - \frac{1}{1000} \right] = 1,63 \cdot 10^{-3} \text{ kg} \cdot \text{mol}^{-1} \quad \text{2 mark}$$



$$\Delta H^\circ_{\text{vap}}(\text{H}_2\text{O}) = -241,8 - (-286,8) = 44,0 \text{ kJ} \cdot \text{mol}^{-1} \quad \text{2 mark}$$

$$\Delta S^\circ_{\text{vap}}(\text{H}_2\text{O}) = \frac{\Delta H^\circ_{\text{vap}}(\text{H}_2\text{O})}{T_{\text{vap}}} = \frac{42000}{373,15} = 113 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \quad \text{2 mark}$$

$\Delta S^\circ_{\text{fus}}$ can also be calculated by subtracting the absolute entropies of the gas and liquid at 100 °C.

13.3. Using the above results and the Clausius-Clapeyron equations for state changes, calculate the melting temperatures T_{fus} and boiling temperatures T_{eb} (in °C) of water at 74.7 kPa.

$$\frac{P - P^\circ}{T_{\text{fus}} - T^\circ_{\text{fus}}} = \frac{\Delta H^\circ_{\text{fus}}}{\Delta V^\circ_{\text{fus}}} \rightarrow \frac{74,7 - 101,3}{T_{\text{fus}} - 273,15} = \frac{6000}{1,63 \cdot 10^{-3}} \rightarrow \Delta T_{\text{fus}} = -6,9 \cdot 10^{-6} \text{ K}$$

$$T_{\text{fus}} \approx T^\circ_{\text{fus}} = 273,15 \text{ K} = 0 \text{ °C} \quad \text{2 mark}$$

$$\ln\left(\frac{P}{P^\circ}\right) = -\frac{\Delta H^\circ_{\text{vap}}}{R} \cdot \left(\frac{1}{T_{\text{eb}}} - \frac{1}{T^\circ_{\text{eb}}}\right) \rightarrow \ln\left(\frac{74,7}{101,3}\right) = -\frac{44000}{8,314} \cdot \left(\frac{1}{T_{\text{eb}}} - \frac{1}{373,15}\right)$$

$$T_{\text{eb}} = 365 \text{ K} = 92 \text{ °C} \quad \text{2 mark}$$

The previous results, therefore, do not justify the existence of ice under these conditions. The students became more thoughtful when they noticed that the weather station on top of the mountain indicated 100 kPa, 5.0 °C and a relative humidity of 80%.

13.4. To be sure of all his reasoning, Caesar calculated the spontaneity of the ice-melting process under these conditions: what result did he obtain?

You must calculate $\Delta G^\circ_{\text{fus}}$ (5 °C) in kJ/mol.



$$\Delta G^\circ_{\text{fus}}(\text{H}_2\text{O}) = \Delta H^\circ_{\text{fus}}(\text{H}_2\text{O}) - T \cdot \Delta S^\circ_{\text{fus}}(\text{H}_2\text{O})$$

$$\Delta G^\circ_{\text{fus}}(\text{H}_2\text{O}) = 6,0 - (5 + 273,15) \cdot 22,0 \cdot 10^{-3} = -0,12 \text{ kJ} \cdot \text{mol}^{-1} < 0$$

The process is spontaneous and therefore, the ice melts at this temperature. **3 mark**

Then it occurred to Erik that perhaps there was no water in the mountains because it was evaporating. Just at that moment, a strong wind began to blow, the clouds loaded with moisture collided with the mountains, and it started to rain, even from the bottom up, a product of the orography. He realized that for condensation to occur at a given temperature, the air must first be saturated with water vapor; the relative humidity must be $H_r = 100\%$ inside the cloud.

13.5. Calculate the pressure of water vapor at 5.0°C when the air is saturated with moisture $P_{\text{H}_2\text{O}}^{\text{sat}}$ in kPa.



$$\Delta G^\circ_{\text{vap}}(\text{H}_2\text{O}) = \Delta H^\circ_{\text{vap}}(\text{H}_2\text{O}) - T \cdot \Delta S^\circ_{\text{vap}}(\text{H}_2\text{O})$$

$$\Delta G^\circ_{\text{vap}}(\text{H}_2\text{O}) = 44,0 - (5 + 273,15) \cdot 113 \cdot 10^{-3} = 12,6 \text{ kJ} \cdot \text{mol}^{-1} < 0 \quad \text{2 mark}$$

$$\Delta G^\circ_{\text{vap}}(\text{H}_2\text{O}) = -R \cdot T \cdot \ln p_r^{\text{sat}}(\text{H}_2\text{O}) \rightarrow 12,6 \cdot 10^3 = -8,314 \cdot 278,15 \cdot \ln p_r^{\text{sat}}(\text{H}_2\text{O}) \quad \text{2 mark}$$

$$p_r^{\text{sat}}(\text{H}_2\text{O}) = 4,36 \cdot 10^{-3} \rightarrow p^{\text{sat}}(\text{H}_2\text{O}) = 4,36 \cdot 10^{-3} \cdot 100 = 0,436 \text{ kPa} \quad \text{2 mark}$$

13.6. Help Erik calculate how much water gas (in g) is contained in 10 km^3 of air around Mount Sántis when the relative humidity is 80% at 5.0°C . Consider $P_{\text{H}_2\text{O}}^{\text{sat}} = 0.50 \text{ kPa}$ if he failed to solve **13.5**.

$$p(\text{H}_2\text{O}) = p^{\text{sat}}(\text{H}_2\text{O}) \cdot H_r = 0,436 \cdot 0,80 = 0,349 \text{ kPa}$$

$$n(\text{H}_2\text{O}) = \frac{p(\text{H}_2\text{O}) \cdot V}{R \cdot T} = \frac{0,349 \cdot 10^3 \cdot 10 \cdot (10^3)^3}{8,314 \cdot 278,15} = 1,51 \cdot 10^9 \text{ mol}$$

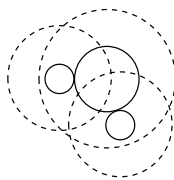
$$m(\text{H}_2\text{O}) = M(\text{H}_2\text{O}) \cdot n(\text{H}_2\text{O}) = 18,02 \cdot 1,51 \cdot 10^9 = 2,72 \cdot 10^{10} \text{ g} = 27 \text{ kton}$$

Using the value $P_{\text{H}_2\text{O}}^{\text{sat}} = 0,50 \text{ kPa}$, the result is $1,73 \cdot 10^9 \text{ g}$ (17 kton) instead.

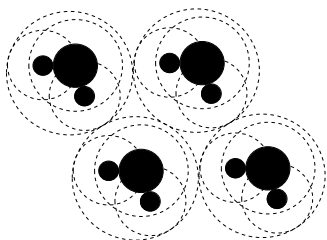
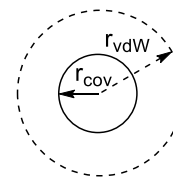
13.7. Condensation also requires a nucleus or surface where the molecules that “escape” from the gas phase are deposited. How much water per unit area is contained in a monolayer (layer one molecule thick) covering the condensation nucleus? Assume that the water molecules are spheres of radius $r = 179 \text{ pm}$ and are arranged as compactly as possible.

Effective radius of a water molecule in constant motion

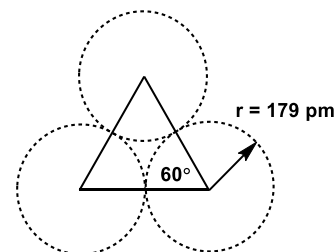
$$r_{\text{H}_2\text{O}}^{\text{eff}} \approx \frac{r_{\text{H}}^{\text{cov}} + r_{\text{H}}^{\text{vdW}} + r_{\text{O}}^{\text{cov}} + r_{\text{O}}^{\text{vdW}}}{2} = 179 \text{ pm}$$



Covalent and van der Waals radii



Arrangement of water molecules in a compact monolayer



Assuming that water molecules occupy the entire surface

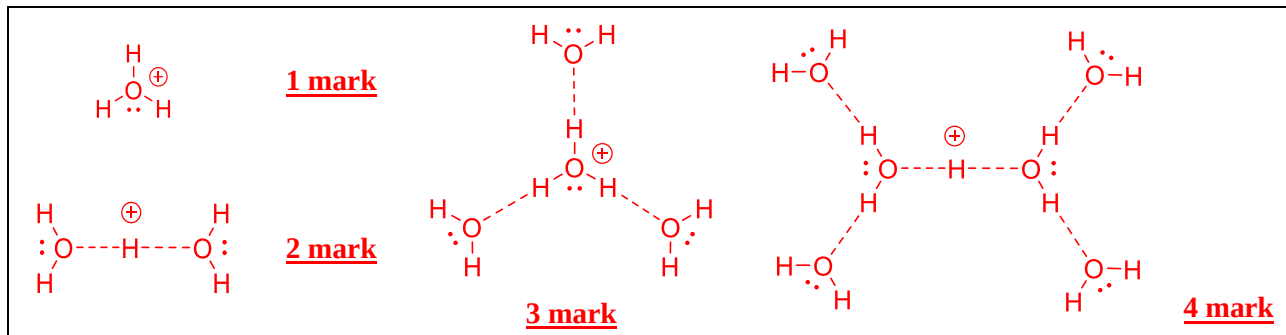
$$N_{\text{H}_2\text{O}}^{\text{max}} = \frac{A_{\text{total}}}{A_{\text{molecule}}^{\text{H}_2\text{O}}} = \frac{1 \text{ m}^2}{\pi \cdot (179 \cdot 10^{-12})^2 \text{ m}^2} = 9,93 \cdot 10^{18} \text{ molecules} \quad \text{3 mark}$$

En realidad, solo una fracción está ocupada

$$\frac{A_{\text{ocupada}}^{\text{molecule}}}{A_{\text{triangle}}} = \frac{3 \cdot A_{\text{circular sector}}}{A_{\text{triangle}}} = \frac{3 \cdot \frac{60}{360} \cdot \pi \cdot r^2}{\frac{1}{2} \cdot (2 \cdot r) \cdot \left(\frac{\sqrt{3}}{2} \cdot 2 \cdot r\right)} = 0,907 \quad \text{4 mark}$$

$$N_{\text{H}_2\text{O}} = 0,907 \cdot 9,93 \cdot 10^{18} = 9,01 \cdot 10^{18} \text{ molecules/m}^2$$

13.8. During his search, Erik found a curious fact in the chapter on chemical equilibrium: “In acidic aqueous solution, hydronium ions (H_3O^+), the Zundel cation (H_5O_2^+ , $\text{H}_3\text{O}^+\cdot\text{H}_2\text{O}$, $\text{H}^+\cdot 2\text{H}_2\text{O}$), the Eigen cation (H_9O_4^+ , $\text{H}_3\text{O}^+\cdot 3\text{H}_2\text{O}$) and the Reed-Stoyanov cation ($\text{H}_{13}\text{O}_6^+$, $\text{H}^+\cdot 6\text{H}_2\text{O}$) are present”. Although this did not help solve his questions, he represented the structures of these four species for fun, marking the relevant hydrogen bonds with dashed lines. Can you do it, too? Hint: All structures have some element of symmetry.



13.9. César likes to experiment, and at lunchtime, he poured some of the contents of the salt shaker out the window. When he came out, he found that part of the ice outside had melted, exposing a portion of the ground. What temperature is required for water to freeze again? A saturated sodium chloride solution in water at $5.0\text{ }^\circ\text{C}$ contains 26.3% by mass. $M(\text{NaCl}) = 58.44 \text{ g/mol}$

$$K_{\text{cryo}} = \frac{R \cdot M_{\text{dis}} \cdot T_{\text{fus}}^2}{1000 \cdot \Delta H_{\text{fus}}^\circ}$$

$$K_{\text{cryo}} = \frac{R \cdot M_{\text{dis}} \cdot T_{\text{fus}}^2}{1000 \cdot \Delta H_{\text{fus}}^\circ} = \frac{8,314 \cdot 18,02 \cdot 273,15^2}{1000 \cdot 6000} = 1,86\text{ }^\circ\text{C} \cdot \text{mol} \cdot \text{kg}^{-1} \quad \text{2 mark}$$

$$b(\text{NaCl}) = \frac{\omega_{\%} \cdot 10^3}{(100 - \omega_{\%}) \cdot M(\text{NaCl})} = \frac{26,3 \cdot 10^3}{(100 - 26,3) \cdot 58,44} = 6,11 \text{ mol} \cdot \text{kg}^{-1} \quad \text{2 mark}$$

$$\Delta T_{\text{cong}} = i \cdot b(\text{NaCl}) \cdot K_{\text{cryo}} = 2 \cdot 6,11 \cdot 1,86 \approx 23\text{ }^\circ\text{C} \quad \text{1 mark}$$

$$T_{\text{fus}} = T_{\text{fus}}^\circ - \Delta T_{\text{cong}} = 0 - 23 = -23\text{ }^\circ\text{C} \quad \text{1 mark}$$

13.10. While reflecting, César remembered his visit to the Faculty of Chemistry and the debates with the professors of Thermodynamics and Kinetics, José Manuel Nieto and Rubén Álvarez and at that time, the mystery was solved! Can you infer why there is ice on Mount Säntis in summer?

The slight temperature difference between ice and air causes heat transfer to occur slowly and fusion to take a long time to occur. 2 mark

Question 14. Tasty electrochemistry! (15% of the total; 11th and 12th grades)

14.1	14.2	14.3	14.4	14.5	14.6	14.7	14.8	14.9	14.10	14.11	14.12	Total	Total
4	3	2	5	5	3	7	9	8	5	5	4	60	15%
4	3	2	5	5	3	7	9	8	5	5	4	60	15%

One morning, the Ibero-American contestants Mario and Héctor studied various electrochemical systems. They had a silver chloride reference electrode (AgCl/Ag , $E^\circ = 0.222 \text{ V}$) and a digital multimeter. They decided to measure the potential of another electrode that had been prepared by the laboratory professor using mercury, mercurous sulfate, and a saturated solution of potassium sulfate since they could not find the value in the library books.

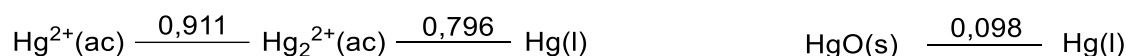
14.1. The electrochemical cell formed by the union of both electrodes had an EMF of 393 mV, with the silver chloride electrode acting as the anode. Represent the thermodynamic notation of the cell and calculate the unknown electrode potential $E[\text{Hg}_2\text{SO}_4(\text{s})/\text{Hg}(\text{l}), \text{SO}_4^{2-}(\text{ac})]$.

$\text{Ag}(\text{s}) \mid \text{AgCl}(\text{s}), \text{Cl}^-(\text{ac}) \parallel \text{SO}_4^{2-}(\text{ac}), \text{Hg}_2\text{SO}_4(\text{s}), \text{Hg}(\text{l}) \mid \text{Pt}(\text{s})$ **2 mark**

$$\Delta E = E(\text{Hg}_2^{2+}/\text{Hg}) - E(\text{AgCl}/\text{Ag}, \text{Cl}^-)$$

$$0,393 = E(\text{Hg}_2^{2+}/\text{Hg}) - 0,222 \rightarrow E(\text{Hg}_2^{2+}/\text{Hg}) = 0,615 \text{ V} \quad \textbf{2 mark}$$

14.2. The presence of the mercurous ion caught Mario's attention, who believed that this ion was not stable in an aqueous solution. He demonstrates that the previous reasoning is incorrect, using the Latimer diagram of mercury.



$$\Delta E_{\text{desp}}^\circ = E^\circ(\text{Hg}_2^{2+}/\text{Hg}) - E^\circ(\text{Hg}^{2+}/\text{Hg}_2^{2+}) = 0,796 - 0,911 = -0,115 \text{ V} < 0 \quad \textbf{2 mark}$$

Therefore, the process is not spontaneous and the disproportionation does not occur. **1 mark**

14.3. Mario was convinced that the mercury(I) concentration had to be very small and that mercurous sulfate should be poorly soluble in water. How could you prove this hypothesis simply in the laboratory?

A portion of the salt and water is added to a test tube. If the disappearance of the solid is not seen when stirring, it can be assumed that the salt is poorly soluble. **2 mark**

14.4. Héctor then found a note from the professor that the electrode had been prepared using a saturated solution of potassium sulfate, which has 10.0% salt by mass and a density of 1.08 g/cm³. How can the solution be prepared and what is its molar concentration? $M(\text{K}_2\text{SO}_4) = 174,27 \text{ g/mol}$.

The solution can be prepared by adding an excess of potassium sulfate to a given volume of water, then stirring and finally filtering the resulting mixture to remove the undissolved solid. **2 mark**

$$c(\text{K}_2\text{SO}_4) = \frac{w\% \cdot d_D \cdot 10}{M(\text{K}_2\text{SO}_4)} = \frac{10 \cdot 1,08 \cdot 10}{174,27} = 0,620 \text{ mol} \cdot \text{L}^{-1} \quad \textbf{3 mark}$$

14.5. Then, Héctor decided to check the previous result. He poured a 5.00 mL aliquot of a freshly prepared saturated potassium sulfate solution into a beaker and added an excess concentrated barium chloride solution. After letting it rest, he filtered out all the white solid formed, dried it and determined that its mass was 720 mg. What is the new concentration value?

$$M(\text{BaSO}_4) = 137,33 + 32,07 + 4 \cdot 16,00 = 233,4 \text{ g} \cdot \text{mol}^{-1} \quad \underline{\text{1 mark}}$$

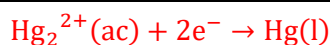
$$n(\text{K}_2\text{SO}_4) = n(\text{SO}_4^{2-}) = n(\text{BaSO}_4) = \frac{m(\text{BaSO}_4)}{M(\text{BaSO}_4)} = \frac{720 \cdot 10^{-3}}{233,4} = 3,08 \cdot 10^{-3} \text{ mol} \quad \underline{\text{2 marca}}$$

$$c(\text{K}_2\text{SO}_4) = \frac{n(\text{K}_2\text{SO}_4)}{V(\text{K}_2\text{SO}_4)} = \frac{3,08 \cdot 10^{-3}}{0,005} = 0,617 \text{ mol} \cdot \text{L}^{-1} \quad \underline{\text{2 marca}}$$

14.6. Mario observed that a tray with elemental sulfur had been placed on the work table. This is a standard procedure to collect accidentally spilled mercury residue. Because?

Mercury and sulfur are soft species that react with each other, forming mercury sulfide, a very thermodynamically stable compound. 3 mark

14.7. After examining all the data, Mario calculated the concentration of mercury(I) ions in the solution and the solubility product constant of mercurous sulfate. What values did he obtain? If I cannot solve **14.1**, consider that the potential of the mercury electrode was 0.620 V.



$$E(\text{Hg}_2\text{SO}_4/\text{Hg}) = E^\circ(\text{Hg}_2^{2+}/\text{Hg}) - \frac{0,0592}{2} \cdot \log \frac{1}{[\text{Hg}_2^{2+}]} \quad \underline{\text{2 mark}}$$

$$0,615 = 0,796 - \frac{0,0592}{2} \cdot \log \frac{1}{[\text{Hg}_2^{2+}]} \rightarrow [\text{Hg}_2^{2+}] = 7,68 \cdot 10^{-7} \text{ mol} \cdot \text{L}^{-1} \quad \underline{\text{3 mark}}$$

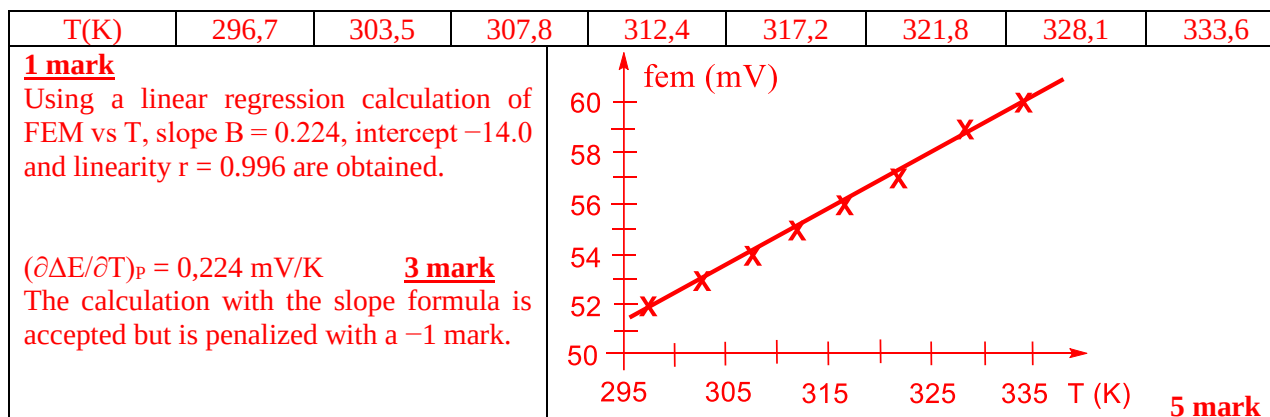
$$K_{\text{ps}}(\text{Hg}_2\text{SO}_4) = [\text{Hg}_2^{2+}] \cdot [\text{SO}_4^{2-}] = 7,68 \cdot 10^{-7} \cdot 0,615 = 4,76 \cdot 10^{-7} \quad \underline{\text{2 mark}}$$

There was also cadmium in the laboratory, and the students decided to study the formation reaction of an amalgam of this metal, for which they measured the EMF of a new cell.

$\text{Pt(s)}|\text{Cd(s)}|\text{CdSO}_4(\text{ac}, 0,1 \text{ mol/L})|\text{Cd(Hg)}|\text{Pt(s)}$

T(°C)	23,5	30,3	34,6	39,2	44,0	48,6	54,9	60,4
FEM (mV)	52	53	54	55	56	57	59	60

14.8. Using the data provided, complete the ΔE vs T graph and calculate the cell temperature coefficient $(\partial \Delta E / \partial T)_p$ (the slope of the curve), assuming that the relationship between the variables is linear.



14.9. Calculate the value of the indicated thermodynamic variables at 298.15 K, with two significant figures.

$$\Delta S = -z \cdot F \cdot (\partial \Delta E / \partial T)_p$$

You must calculate ΔE (in mV), ΔG (in kJ/mol), ΔS (in J/(mol·K)) and ΔH (in kJ/mol). T = 298,15 K



For this reaction $z = 2$ since a Cd^{2+} solution is present at each electrode.

$$\Delta S_r = -z \cdot F \cdot (\partial \Delta E / \partial T)_P = -2 \cdot 96485 \cdot 0,224 \cdot 10^{-3} = -43,2 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} \quad \underline{\text{2 mark}}$$

$$\Delta E_r = (\partial \Delta E / \partial T)_P \cdot T + \text{intercept} = 0,224 \cdot 298,15 - 14,0 \approx 52 \text{ mV} \quad \underline{\text{2 mark}}$$

$$\Delta G_r = -z \cdot F \cdot \Delta E_r = -2 \cdot 96485 \cdot 0,052 \cdot 10^{-3} = -10,0 \text{ kJ} \cdot \text{mol}^{-1} \quad \underline{\text{2 mark}}$$

$$\Delta H_r = \Delta G_r + T \cdot \Delta S_r = -10,0 + 298,15 \cdot (-43,2 \cdot 10^{-3}) = -22,9 \text{ kJ} \cdot \text{mol}^{-1} \quad \underline{\text{2 mark}}$$

After work, the students drank carbonated soda. While they were resting, Héctor wondered why, if soda cans are made of aluminum, an active metal, no chemical reaction occurs. This question made them search the Internet, where they found the information they needed.

14.10. The typical acidulant in carbonated soft drinks is phosphoric acid. What is the acid concentration capable of giving the drink an acidity of $\text{pH} = 3.0$? $\text{pK}_{a1} = 2,15$; $\text{pK}_{a2} = 7,20$; $\text{pK}_{a3} = 12,35$

It is possible to neglect the second and third dissociation.

$$[\text{H}^+] = 10^{-\text{pH}} = 10^{-3} \text{ mol} \cdot \text{L}^{-1} \quad \underline{\text{1 mark}}$$

The acid concentration at equilibrium differs considerably from the initial concentration since K_{a1} is relatively large and also $\text{pH} > \text{K}_{a1}$. Then $\underline{\text{1 mark}}$

$$\text{K}_{a1} = \frac{[\text{H}^+][\text{H}_2\text{PO}_4^-]}{[\text{H}_3\text{PO}_4]} = \frac{[\text{H}^+]^2}{c_0(\text{H}_3\text{PO}_4) - [\text{H}^+]}$$

$$c_0(\text{H}_3\text{PO}_4) = 10^{-3} + \frac{(10^{-3})^2}{10^{-2,15}} = 1,141 \cdot 10^{-3} \text{ mol} \cdot \text{L}^{-1} \quad \underline{\text{3 mark}}$$

14.11. Show that the reaction between metallic aluminum and the acid in the soft drink is spontaneous at room temperature. $E^\circ(\text{Al}^{3+}/\text{Al}) = -1.66 \text{ V}$. You must calculate ΔE (in V) and ΔG (in kJ/mol)



$$E(\text{H}^+/\text{H}_2) = -0,0592 \cdot \text{pH} = -0,0592 \cdot 3 = -0,18 \text{ V} \quad \underline{\text{2 mark}}$$

$$\Delta E = E(\text{H}^+/\text{H}_2) - E^\circ(\text{Al}^{3+}/\text{Al}) = -0,18 - (-1,66) = 1,48 \text{ V} > 0 \quad \underline{\text{2 mark}}$$

So, the process is spontaneous. $\underline{\text{1 mark}}$

14.12. In reality, the aluminum in soda cans is covered with a layer of complex composition that protects it against oxidation due to acid solution. What maximum value can the dissolved aluminum concentration take within the layer? $\text{pK}_{ps}[\text{Al}(\text{OH})_3] = 32.97$.

Solution 1: Since there is no solution within the oxide layer, it can be assumed that

$$[\text{OH}^-] = 3 \cdot [\text{Al}^{3+}] \rightarrow \text{K}_{ps}(\text{Al}(\text{OH})_3) = [\text{Al}^{3+}] \cdot [\text{OH}^-]^3 = 27 \cdot [\text{Al}^{3+}]^3 = 10^{-32,97} \quad \underline{\text{2 mark}}$$

$$[\text{Al}^{3+}] = 2,4 \cdot 10^{-9} \text{ mol} \cdot \text{L}^{-1} \quad \underline{\text{2 mark}}$$

Solution 2: However, on the surface of the oxide, in direct contact with the solution, it is true that

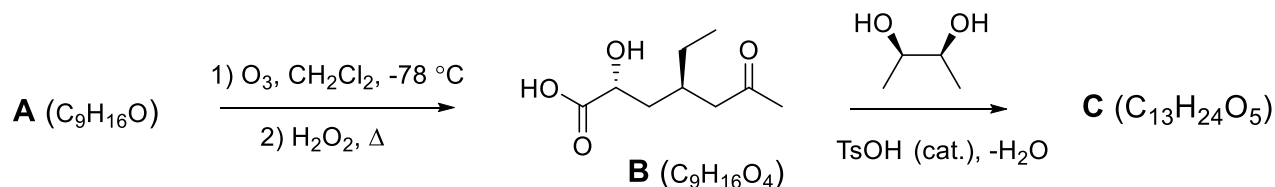
$$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} = \frac{1,0 \cdot 10^{-14}}{10^{-3}} = 1,0 \cdot 10^{-11} \text{ mol} \cdot \text{L}^{-1} \quad \underline{\text{2 mark}}$$

$$[\text{Al}^{3+}] = \frac{\text{K}_{ps}(\text{Al}(\text{OH})_3)}{[\text{OH}^-]^3} = \frac{10^{-32,97}}{[10^{-11}]^3} = 1,07 \text{ mol} \cdot \text{L}^{-1} \quad \underline{\text{2 mark}}$$

Question 15. Play my chemistry in stereo! (12% of the total; only 12th grade)

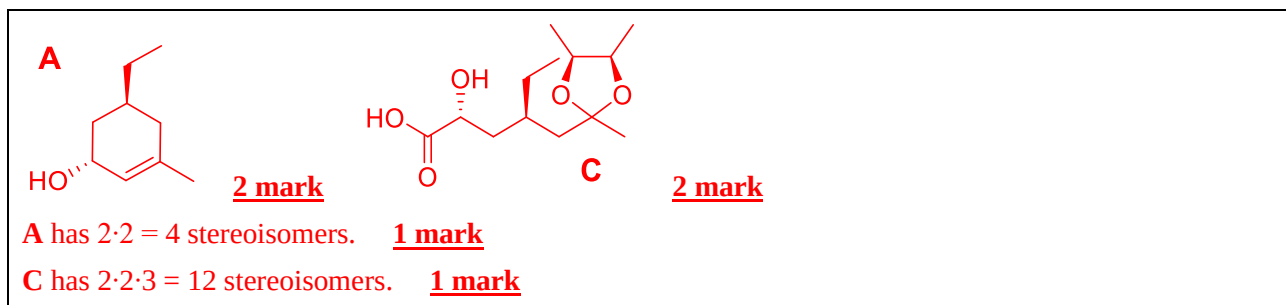
15.1	15.2	15.3	15.4	15.5	15.6	15.7	15.8	15.9	15.10	Total	Total
6	2	5	5	4	4	4	9	3	6	48	12%
6	2	5	5	4	4	4	9	3	6	48	12%

In the past, chemists widely used ozonolysis to determine the structure of organic compounds. This reaction also allows the formation of a doubly functionalized derivative starting from cyclic structures, such as chiral alcohol **A**, as shown in the following sequence.

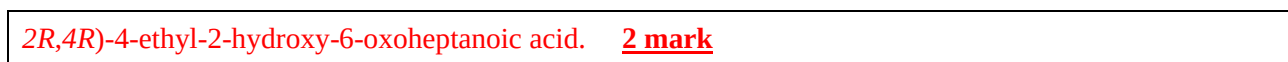


Throughout the question, you must correctly assign the stereochemistry of all centers.

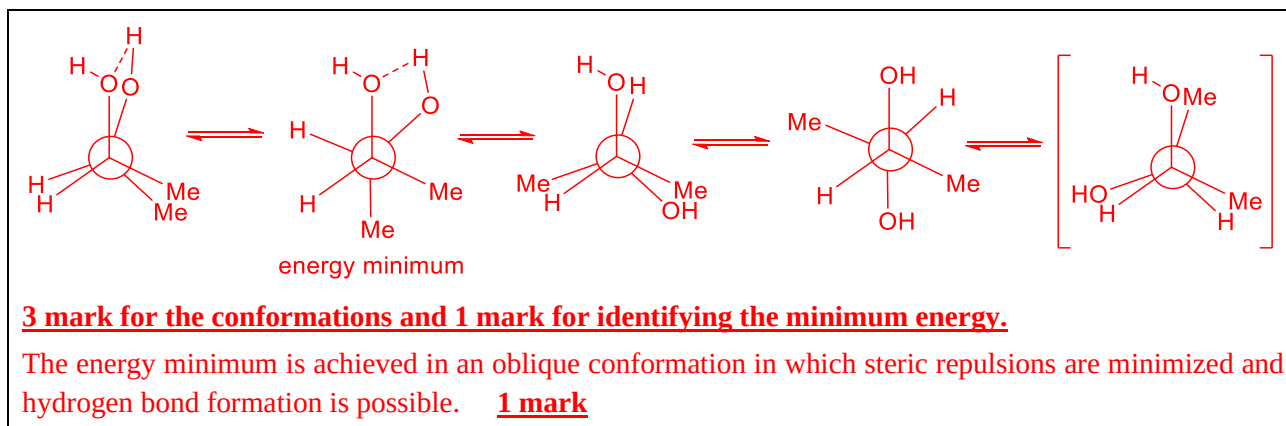
15.1. Represent the structures of **A** and **C**. Tell how many stereoisomers each molecule has.



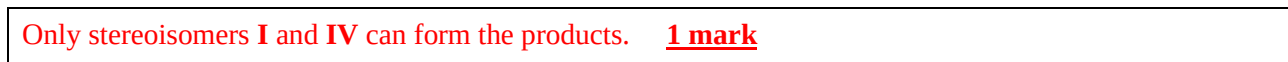
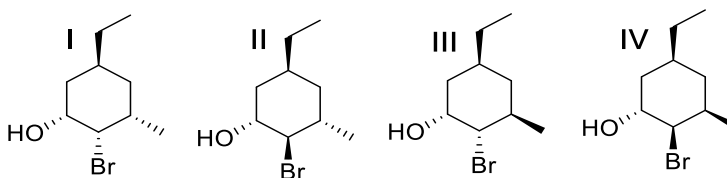
15.2. Name compound **B**.

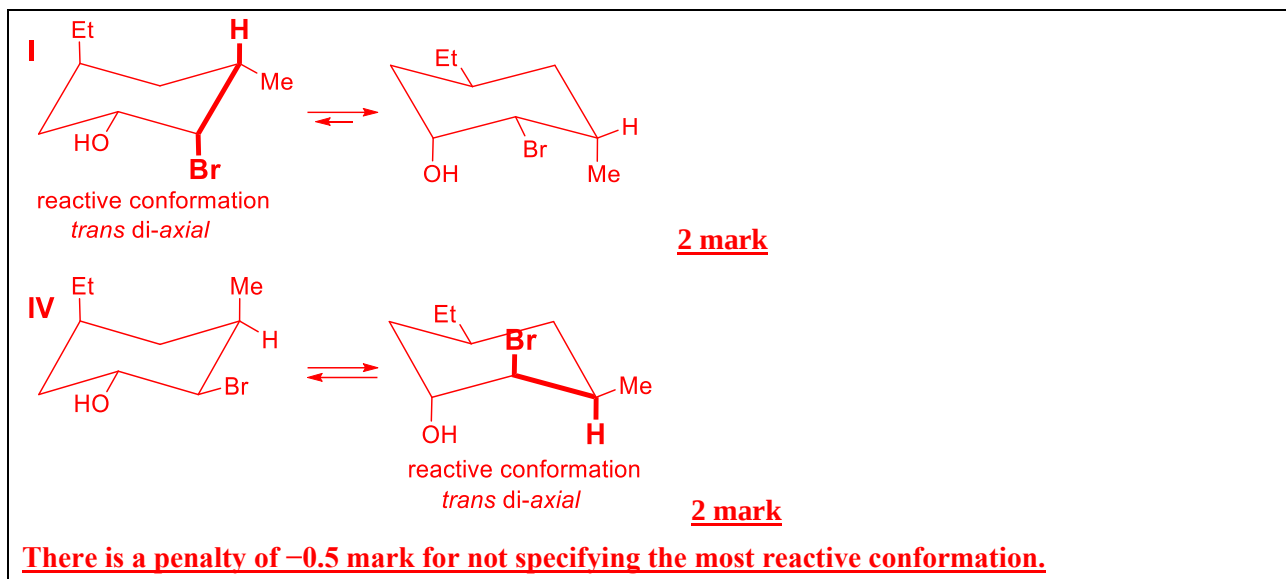


15.3. Represent the most important conformations of *meso*-2,3-butanediol (see structure in the diagram above) using Newman projections. Predict which conformation has the least energy and explain your reasoning.

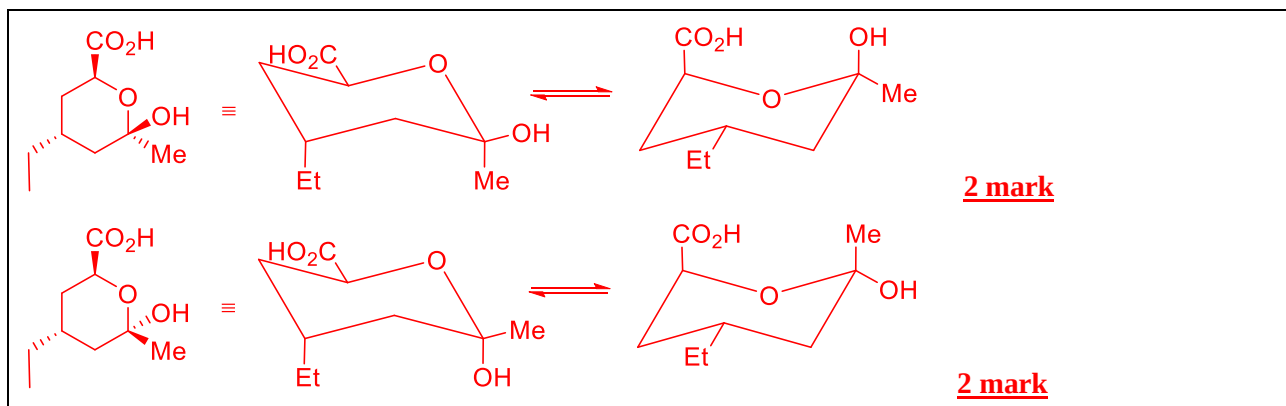


15.4. What two stereoisomers have been shown to generate **A** by bimolecular elimination? Represent the equilibrium of their *chair* conformations and indicate the reactive conformation.

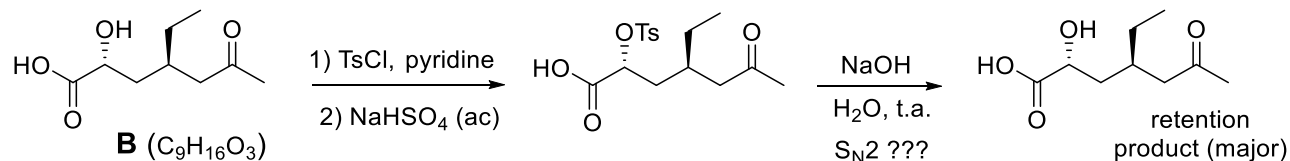




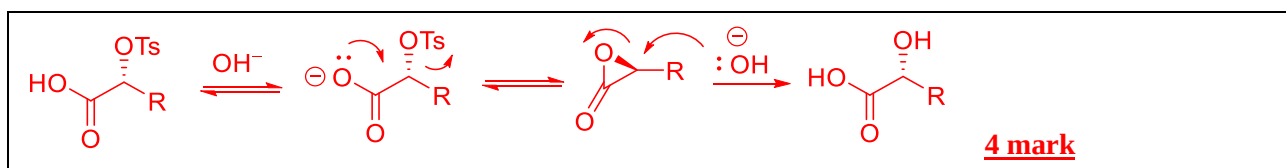
15.5. Ketol **B** exists in the form of two cyclic hemiacetals **B'** and **B''** that are diastereomers of each other. Represent the equilibrium that exists between the hemiketol *chair* conformations.



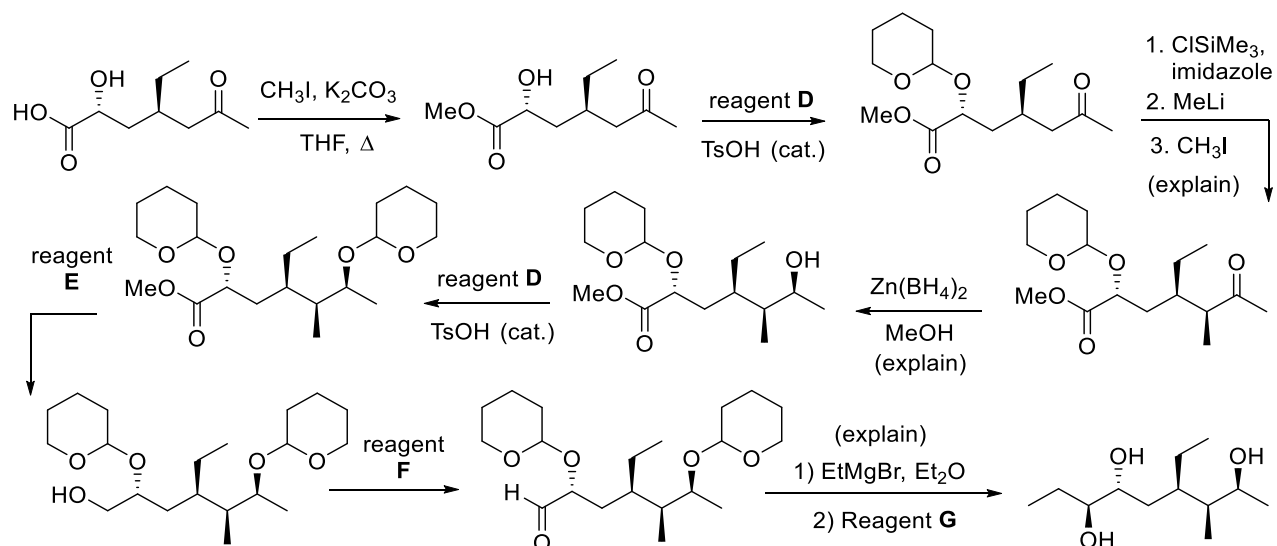
To change the stereochemistry of alcohol **B**, it was treated with tosyl chloride and then sodium hydroxide was added. However, when the product was analyzed, an appreciable amount of the reactant was detected.



15.6. Pose a reaction mechanism that explains the result of the second step above.



Subsequently, the ozonolysis product was subjected to the following sequence of reactions.

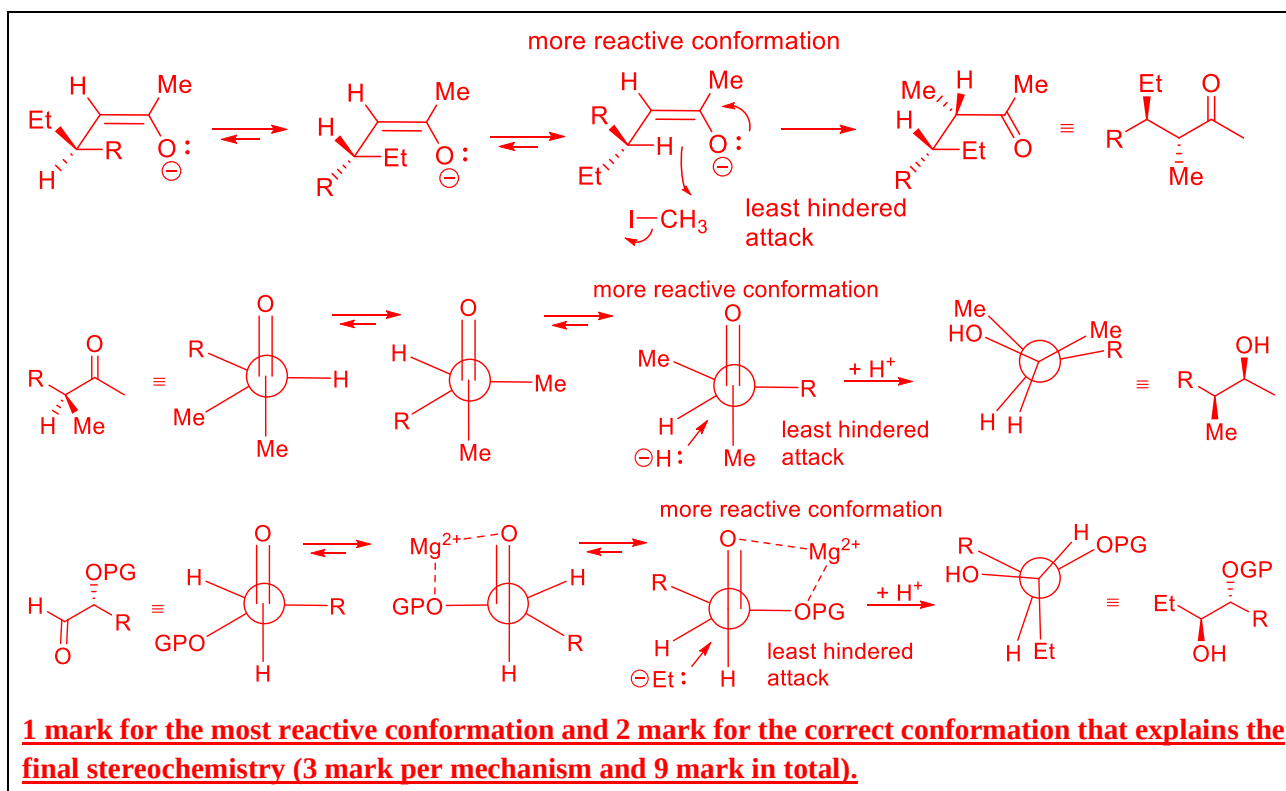


15.7. Suggest what the reagents could be **D**, **E**, **F** and **G**.

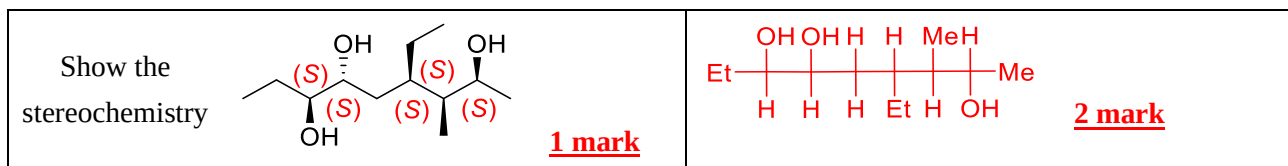
D = DHP (dihhydropyran), **E** = LiAlH_4 / B_2H_6 , **F** = PCC / PDC / Dess-Martin periodinane... **G** = H_3O^+

1 mark for each reagent (4 marks in total).

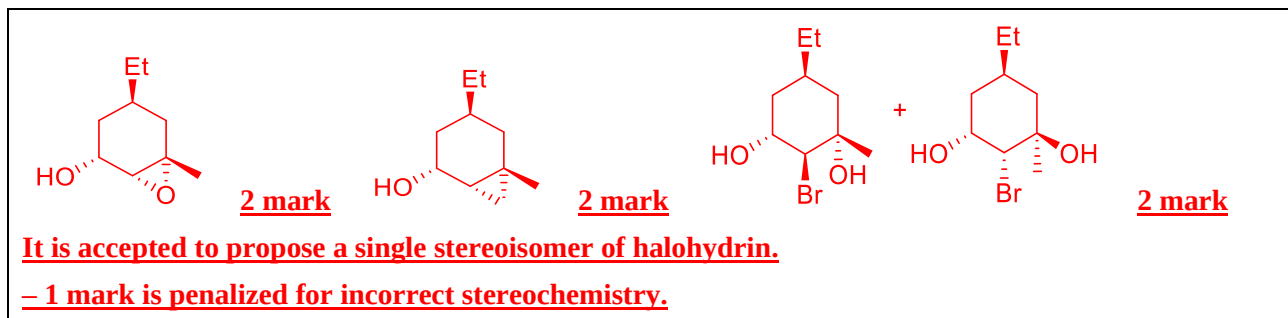
15.8. Propose mechanisms that justify the stereoselectivity of the indicated nucleophilic methylation, reduction and addition steps, using the Felkin-Ahn or Houk models, as appropriate. In your scheme, you only need to represent the conformation of the reactants that give rise to the product.



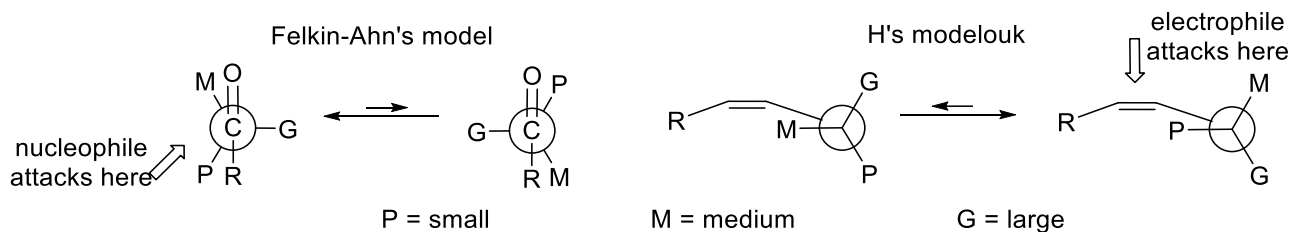
15.9. Represent the Fischer projection of the triol product of the above sequence. Classify the absolute stereochemistry of carbons according to Cahn-Ingold-Prelog.



15.10. Represent the major products of the reaction of compound **A** with: i) AMCPB, ii) $\text{CH}_2\text{I}_2/\text{Zn}(\text{Cu})$ y iii) $\text{Br}_2, \text{H}_2\text{O}$.



TsOH: *para*-toluenesulfonic acid; AMCPB: meta-chloroperoxy-benzoic acid; THF: tetrahydrofuran.

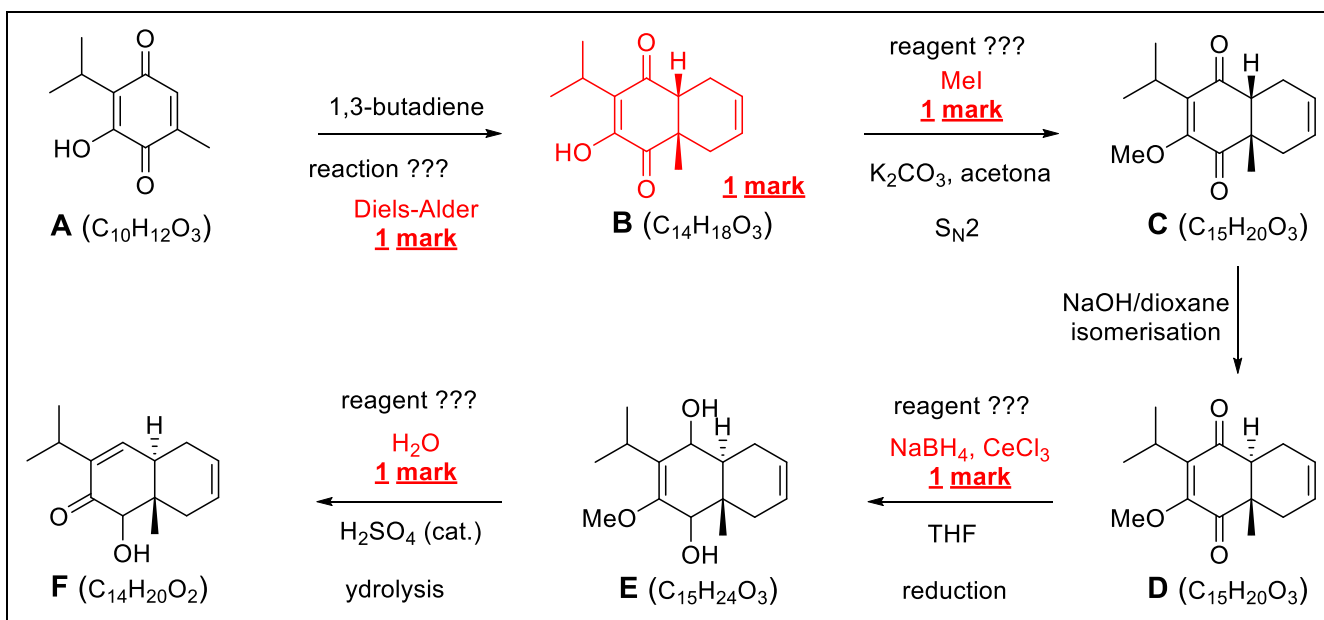


Question 16. Aromatic chemistry (13% of the total; only 12th grade)

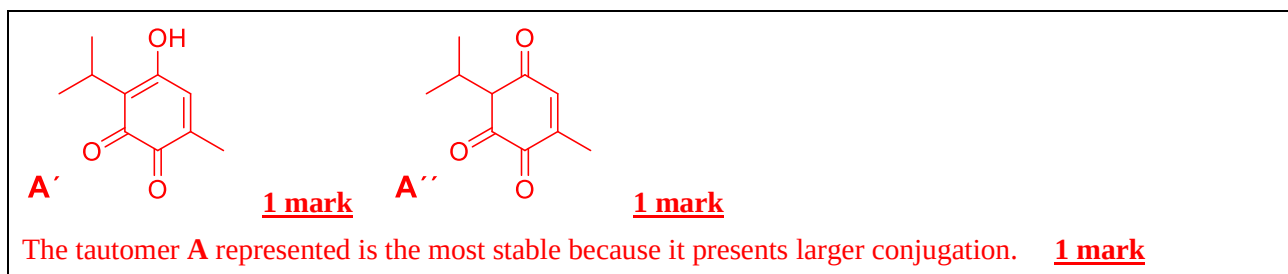
16.1	16.2	16.3	16.4	16.5	16.6	16.7	16.8	16.9	16.10	16.11	16.12	16.13	16.14	Total	Total
5	3	2	1	6	1	5	1	7	4	7	3	4	3	52	13%
5	3	2	1	6	1	5	1	7	4	7	3	4	3	52	13%

When an organic chemist observes the hexagons of wasp or bee honeycombs, he cannot help but think of the benzene ring, the aromatic organic substance par excellence. In this question, several synthetic sequences are shown that allow the obtaining of compounds of this type.

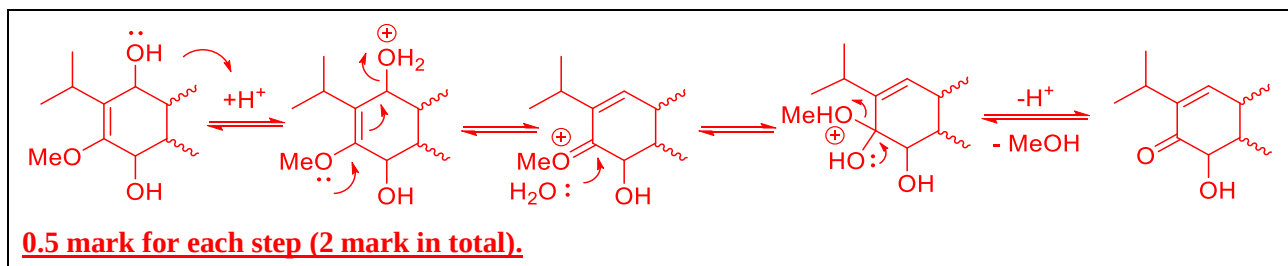
16.1. Fill in the missing reactants, structures, and reaction names in the scheme.



16.2. Compound **A** exists in the form of two other tautomers **A'** and **A''**, represent their structures and say why the tautomer shown is the most stable of all.



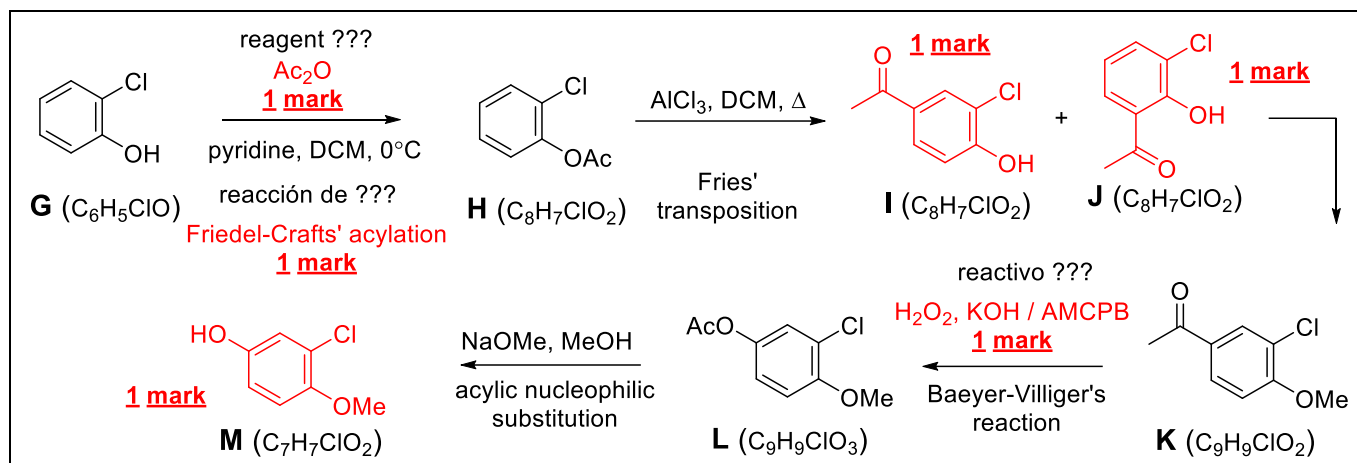
16.3. Propose a detailed mechanism for the hydrolysis reaction of **E** forming **F**.



16.4. Why does the isomerization reaction of *cis* **C** bicyclo occur in *trans* **D**?



16.5. Fill in the missing reactants, structures, and reaction names in the scheme.

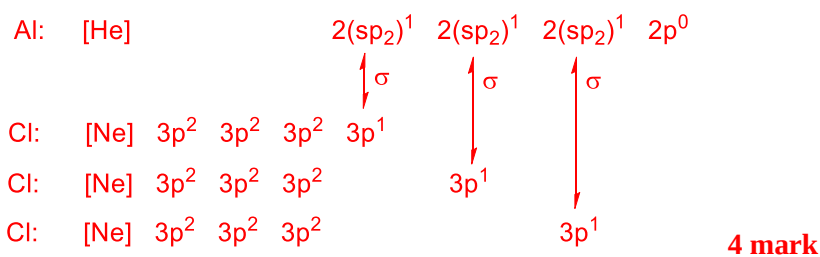


16.6. Name compound G.

o-chlorophenol **1 mark**

16.7. Why aluminum chloride that can act as a catalyst? Represent the electronic distribution of this molecule using hybrid orbitals for the central atom.

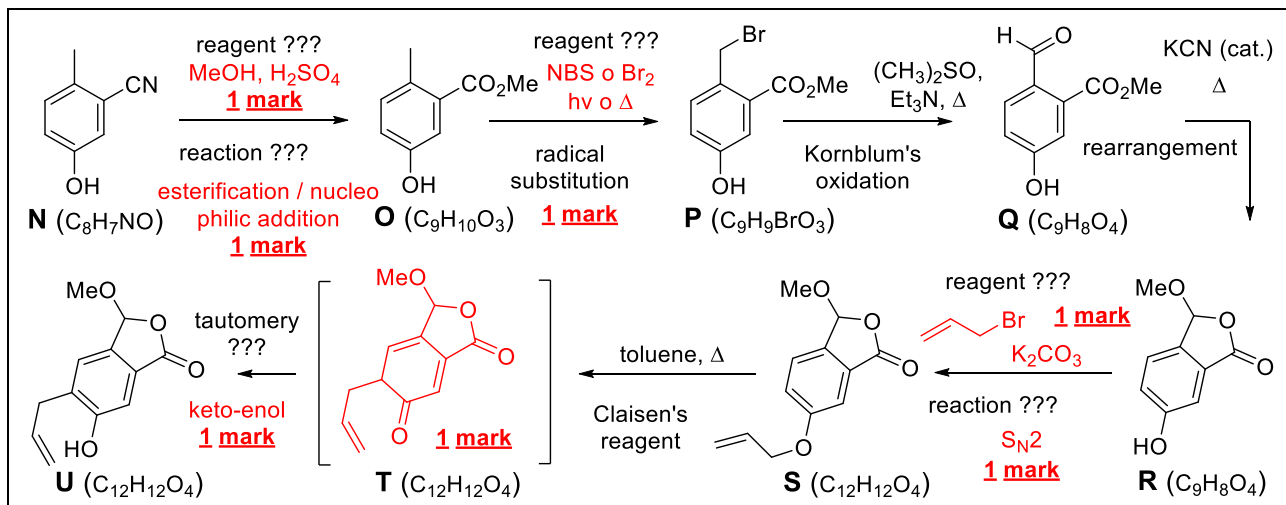
Aluminum has an empty 3p orbital that can accept electrons and act as a Lewis acid. **1 mark**



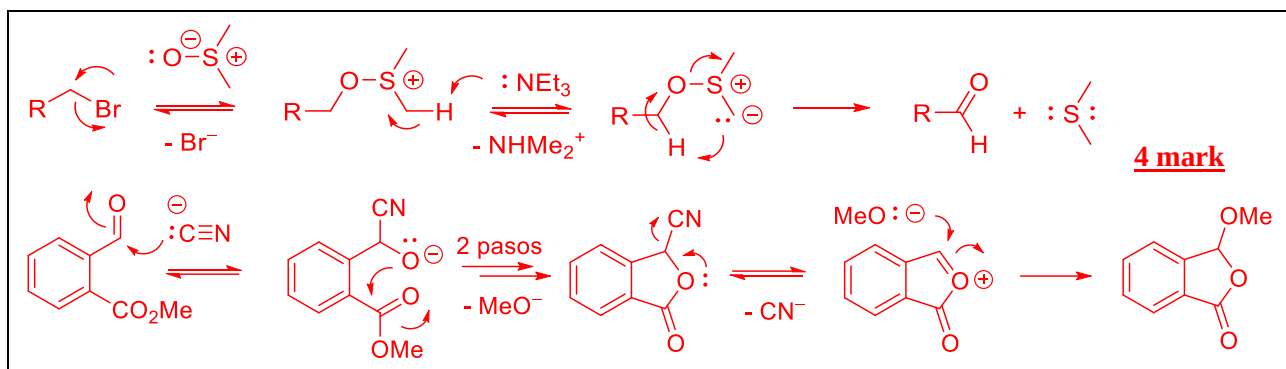
16.8. Compare the acidity of G and M. Explain your reasoning.

G is more acidic than M because it has an electrode group that reduces acidity. **1 mark**

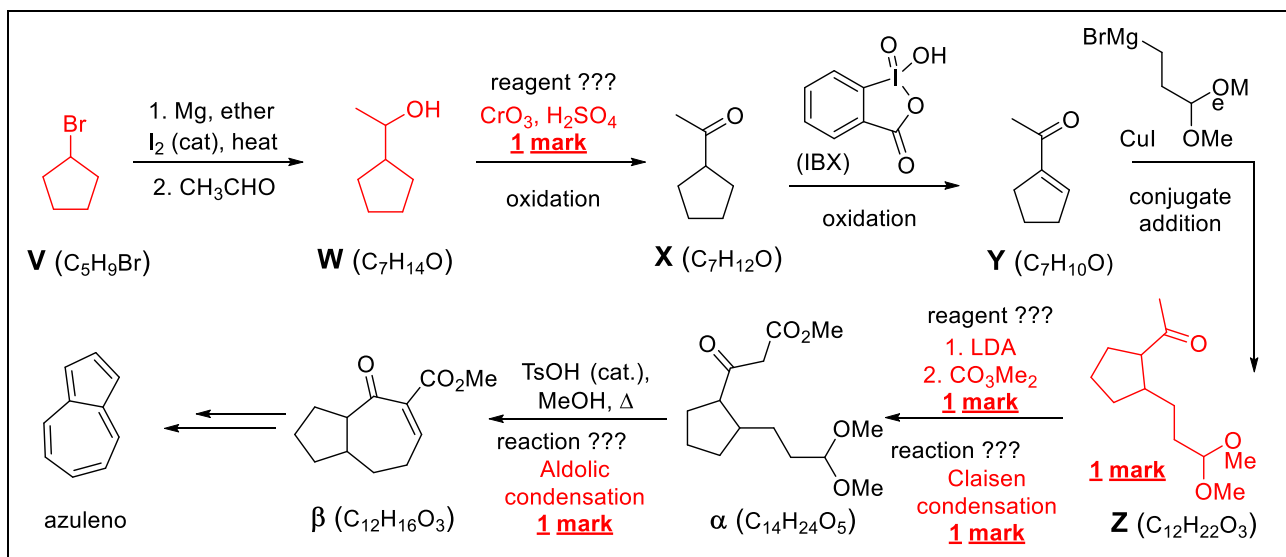
16.9. Fill in the missing reactants, structures, and reaction names in the scheme.



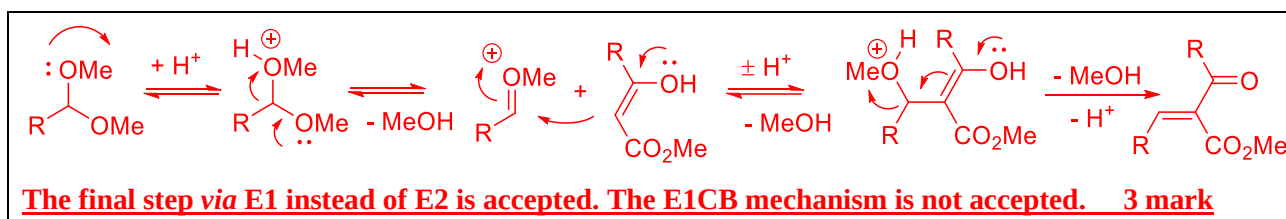
16.10. Propose mechanisms for Kornblum oxidation and cyanide-catalyzed rearrangement.



16.11. Fill in the missing reactants, structures, and reaction names in the scheme.



16.12. Propose a mechanism for the ring closure reaction of **α** in an acidic medium.

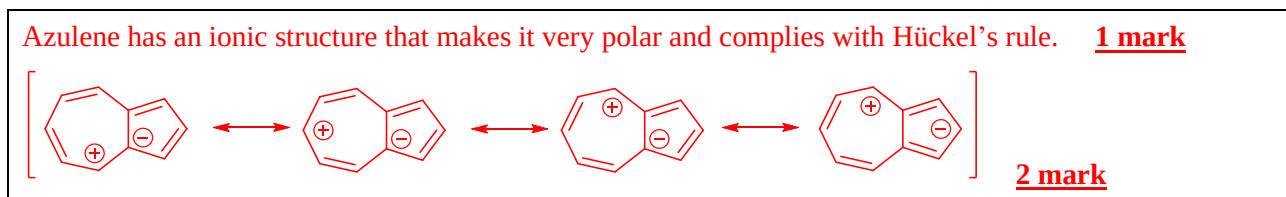


16.13. Propose a sequence of reagents that can ultimately be used to obtain azulene.

1. KOH, H₂O; 2. CaO, NaOH, Δ, -CO₂; 3. Pd, Δ, -H₂O, -H₂

More complex sequences are acceptable, for example, those that do not involve dehydrogenation. **4 mark**

16.14. Why is azulene aromatic and has high polarity? Represent four resonance structures.



//////////////////////**RESERVED QUESTIONS**//////////////////////

When performing an elemental analysis, Daniela and Lianet found that a certain jar with a mass of 85 kg was composed of several mixed micas and other minerals: 19.0% paragonite $\text{NaAl}_2(\text{AlSi}_3)\text{O}_{10}(\text{OH})_2$, 10.8% of zinnwaldite $\text{KLiFeAl}(\text{AlSi}_3)\text{O}_{10}(\text{OH})_2$, 8.2% of phlogopite $\text{KMgMn}_2(\text{AlSi}_3)\text{O}_{10}(\text{OH})_2$, 20.0% of silicon dioxide and 40.0% of kaolin (in mass percent) .

7.4. What mass of silicon is contained in 100 kg of the material that makes up the jar analyzed? Assume that kaolin contains 30.1% silicon by mass.

7.5. If the previous jar is pulverized and calcined in the presence of excess air, by how much does its mass increase? Hint: Only the elements manganese and iron vary their oxidation state.

//////////////////////**RESERVED QUESTIONS**//////////////////////

10.5. Find an expression for the rate of an enzymatic reaction (analogous to Equation 1) using the steady-state approximation for the concentration of the enzyme-substrate complex ES.

Hint: Assume that there is no fast equilibrium for the formation of the complex.

10.6. Under what conditions the expression found by you in **10.4.** is equal to equation (1).